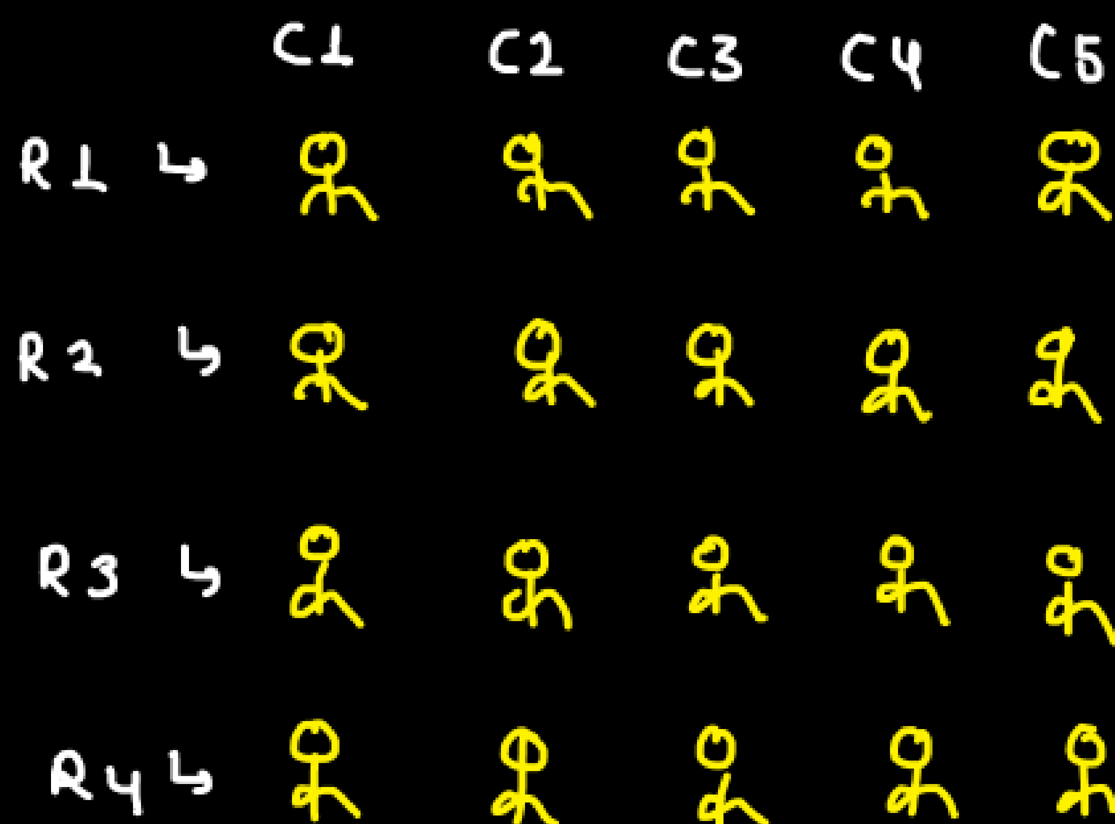
Linear Algebra

↳ Understanding some Basic Terms:-

* Rows and Columns:-



No. of Rows = 4

No. of Columns = 5

Order = Row × column

= 4 × 5

A picture from Morning Assembly

Matrix :-

A rectangular arrangement of elements in rows and columns.

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 4 & 1 & 6 \\ 3 & -1 & \alpha \\ 2 & 0 & 1 \end{bmatrix}_{4\times 3}$$

order = 4×3

Rows = 4

columns = 3

no. of elements = $12 = 4 \times 3$

$$B = \begin{bmatrix} 2 & \alpha & \beta & -1 & 3 \\ 1 & 0 & 16 & \gamma & 1 \end{bmatrix}_{2\times 5}$$

order = 2×5

Rows = 2

columns = 5

no. of elements = $10 = 2 \times 5$

Q. Make a 3x4 Matrix have each element as a odd no.

↳

$$A = \begin{array}{c c} & \begin{array}{cccc} c_1 & c_2 & c_3 & c_4 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array} \\ \begin{array}{c} R_1 \to \\ R_2 \to \\ R_3 \to \end{array} & \begin{bmatrix} 1 & 3 & 7 & 9 \\ 11 & 5 & 21 & 17 \\ 13 & 19 & 27 & 15 \end{bmatrix}_{3\times 4} \end{array}$$

Rows = 3

columns = 4

elements = 3×4 = 12

: no. of

$1 \to 1^{st}$ Row and 1^{st} column $\Rightarrow 1 = a_{11}$

$21 \to 2^{nd}$ Row and 3^{rd} column $\Rightarrow 21 = a_{23}$

$19 \to 3^{rd}$ Row and 2^{nd} column $\Rightarrow 19 = a_{32}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3\times 4}$$

4 **Defining a Matrix:-**

Rows = m

columns = n

Matrix of order = $\underline{m} \times \underline{n}$

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m \times n} = [a_{ij}]_{m \times n}$$

Row

$1 \le i \le m$

$1 \le j \le n$

column

Q.

Question:

Construct a 3×2 matrix whose elements are given by

$$a_{ij} = \frac{1}{2}|i - 3j|.$$

\# Rows = 3

\# column = 2

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}_{3\times 2} = \begin{bmatrix} 1 & 5/2 \\ 1/2 & 2 \\ 0 & 3/2 \end{bmatrix}_{3\times 2}$$

a_{11} : $i=1$, $j=1$; $a_{11} = \frac{1}{2}|1-3| = \frac{1}{2}(2) = 1$

a_{12} : $i=1$, $j=2$; $a_{12} = \frac{1}{2}|1-6| = 5/2$

a_{21} : $i=2$, $j=1$; $a_{21} = 1/2\,|2-3| = 1/2$

↳ Special Types of Matrices:-

1. Rectangular Matrix:- order $m \times n$; where $m \neq n$

$$A = \begin{bmatrix} 1 & 2 \\ -1 & \alpha \\ 0 & 1 \end{bmatrix}_{3\times 2} \qquad B = \begin{bmatrix} 1 & -1 & \alpha \\ 0 & \alpha & 2 \end{bmatrix}_{2\times 3}$$

2. Square matrix:- order $m \times n$; where $m = n$

$$A = \begin{bmatrix} 1 & 3 & \alpha \\ 2 & 16 & \beta \\ -1 & 4 & 0 \end{bmatrix}_{3\times 3} \; ; \quad B = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 3 & 1 & 7 & \alpha \\ \beta & 1 & 2 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}_{4\times 4}$$

↳ Digonal of a Square Matrix :-

Defined only for Square Matrix

Elements $a_{11}, a_{22}, a_{33}, a_{44}$ - - - - are known as Diagonal elements.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times 3} \qquad B = \begin{bmatrix} 4 & 6 & 7 \\ 3 & -2 & 0 \\ \alpha & j & 1 \end{bmatrix}_{3\times 3}$$

diagonal elements $= a_{11}, a_{22}, a_{33}$

Trace of $A = a_{11} + a_{22} + a_{33}$

Diagonal elements $= 4, -2, 1$

Trace of $B = 4 + (-2) + 1$

$= 3$

3. Diagonal Matrix :-

A diagonal matrix is a square matrix in which all elements outside the main diagonal are zero, and the elements of the main diagonal may or may not be zero.

Eg.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad \begin{bmatrix} 5 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow \text{NOT a Diagonal Matrix}$$

4. Scalar Matrix :-

A scalar matrix is a diagonal matrix in which all the elements of the main diagonal are equal.

$$[A] = [a_{ij}]_{m \times n} = \begin{cases} K ; & i = j \\ 0 ; & i \neq j \end{cases}$$

Eg.

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \checkmark, \quad \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \checkmark, \quad \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \times, \quad \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \times$$

NOT Diagonal = NOT Scalar

$$\boxed{\begin{aligned} & a_{11} = a_{22} = a_{33} = 4 \\ & a_{12} = a_{13} = a_{21} = a_{23} \\ & \quad = a_{31} = a_{32} = 0 \end{aligned}}$$

$2 \neq 0$

Diagonal Matrix ✓

Scalar Matrix ✗

$1 \neq 2$

Diagonal Matrix ✓

Scalar Matrix ✗

Square Matrix → Diagonal Matrix → Scalar → Identity

5. Identity Matrix :-

An identity matrix is a square matrix in which all the elements of the main diagonal are equal to 1 and all the other elements are zero.

Identity matrix of order 3:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3}$$

Identity matrix of order 4:

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$$

Identity matrix of order 2:

$$I_5 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2}$$

6. Null Matrix :-

An Matrix [Rectangular or Square] whose all elements are zero.

Eg. - $[0]_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times2}$; $[0]_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times3}$; $[0]_{2\times3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2\times3}$

7. **Row Matrix :-** One Row, Any number of columns.

Rows = 1

column = n

order = $1 \times n$

Eg. → $[1 \quad 2 \quad 3]_{1\times 3}$; $[1 \quad 2]_{1\times 2}$; $[1 \quad 2 \quad -1 \quad 0 \quad 4 \quad 5 \quad 7]_{1\times 7}$

8. **Column Matrix :-** One Column, Any number of Rows.

Rows = m

columns = 1

order = $m \times 1$

Eg. → $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}_{3\times 1}$; $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{2\times 1}$; $\begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \\ 1 \\ 1 \end{bmatrix}_{6\times 1}$

Row Matrix :- $[\qquad\qquad]_{1\times n}$

Column Matrix :- $\begin{bmatrix} \\ \\ \\ \end{bmatrix}_{m\times 1}$

9. Upper Triangular Matrix :-

An upper triangular matrix is a square matrix in which all the elements below the main diagonal are zero, and the diagonal and the elements above the main diagonal may or may not be zero.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \begin{bmatrix} 1 & 7 & 0 \\ 0 & 0 & 9 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 9 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Square Matrix ✓
→ Upper Triangular Matrix XX

10. Lower Triangular Matrix :-

A lower triangular matrix is a square matrix in which all the elements above the main diagonal are zero, and the diagonal and the elements below the main diagonal may or may not be zero.

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 7 & 0 & 0 \\ 0 & 9 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 9 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 \\ -1 & 0 & 3 & 1 \\ 2 & 1 & 0 & 4 \end{bmatrix}$$

→ NOT A lower Triangular Matrix

↳ Equality of two matrices:-

Two matrices are said to be equal if they are of the same order and their corresponding elements are equal.

Question:

If

$$A = \begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and $A = B$, find the values of x and y.

$a_{11} = x$ $\quad b_{11} = 1$

↳ $\boxed{x = 1}$ $\quad$ ↳ $\boxed{y = 4}$

↳ $a_{12} = 2$ $\quad b_{12} = 2$

↳ $a_{21} = 3$ $\quad b_{21} = 3$

↳ $a_{22} = y$ $\quad b_{22} = 4$

Question:

If

$$A = \begin{bmatrix} x+y & 2 & y-z \\ 3 & x-z & 5 \end{bmatrix}_{2\times 3}, \quad B = \begin{bmatrix} 5 & 2 & 1 \\ 3 & 2 & 5 \end{bmatrix}_{2\times 3}$$

and $A = B$, find the values of x, y, and z.

↳ $x + y = 5$ — (1) $\quad$ ↳ $y - z = 1$ — (2)

$x - y = 1$ $\quad$ ↳ $x - z = 2$ — (3)

$\boxed{x = 3, \; y = 2, \; z = 1}$

$A = B$

Note: Equal, equivalent, and similar matrices are all different concepts with different purposes.

Addition and Subtraction of Matrices:-

Matrix addition and subtraction are defined only for matrices of the same order. In addition, corresponding elements of the matrices are added, and in subtraction, corresponding elements are subtracted.

Question:

If

$$A = \begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix}_{2\times 2}, \quad B = \begin{bmatrix} 1 & z \\ 4 & 5 \end{bmatrix}_{2\times 2}$$

and

$$\begin{bmatrix} x+1 & 2+z \\ 7 & y+5 \end{bmatrix} = A + B = \begin{bmatrix} 5 & 7 \\ 7 & 9 \end{bmatrix},$$

find the values of x, y, and z.

$$B + A = \begin{bmatrix} 1+x & z+2 \\ 7 & 5+y \end{bmatrix}$$

$x + 1 = 5 \Rightarrow x = 4$

$2 + z = 7 \Rightarrow z = 5$

$y + 5 = 9 \Rightarrow y = 4$

$$\boxed{A + B = B + A}$$

Q.

Question:

If

$$A = \begin{bmatrix} x & 2 \\ 3 & y \\ z & 5 \end{bmatrix}_{3\times 2}, \quad B = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}_{3\times 2}$$

and

$$A - B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 4 \end{bmatrix},$$

find the values of x, y, and z.

$$A - B = \begin{bmatrix} x-1 & 1 \\ 1 & y-3 \\ z-4 & 4 \end{bmatrix}$$

$x - 1 = 2 \Rightarrow x = 3$

$y - 3 = 2 \Rightarrow y = 5$

$z - 4 = 3 \Rightarrow z = 7$

$$\boxed{x = 3,\ y = 5,\ z = 7}$$

Properties:

$A + B = B + A$

$(A + B) + C = A + (B + C)$

$A + 0 = A = 0 + A$

$A + (-A) = 0 \rightarrow$ Null Matrix

$$B - A = \begin{bmatrix} 1-x & -1 \\ -1 & 3-y \\ 4-z & -4 \end{bmatrix}$$

$A - B = -(B - A)$

4 Multiplication of a Matrix with a Scalar:-

Scalar multiplication of a matrix means multiplying each element of the matrix by the given scalar. If a matrix $A = [a_{ij}]$ is multiplied by a scalar k, then the resulting matrix is $kA = [k \cdot a_{ij}]$.

Example:

If

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

find $-4A$.

$$-4A = \begin{bmatrix} -4 & -8 & -12 \\ -16 & -20 & -24 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix}_{2\times 4}$$

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & ka_{13} & ka_{14} \\ ka_{21} & ka_{22} & ka_{23} & ka_{24} \end{bmatrix}_{2\times 4}$$

Question:

The trace of a matrix A is -2. Find the trace of the matrix $B = 4A$.

↳ $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ $B = 4A = \begin{bmatrix} 4a_{11} & 4a_{12} \\ 4a_{21} & 4a_{22} \end{bmatrix}$

$Tr(A) = a_{11} + a_{22} = -2$

$Tr(B) = Tr(4A) = 4a_{11} + 4a_{22} = 4[a_{11} + a_{22}]$

$= 4\,Tr(A) = \boxed{-8}$

$Tr(4A) = 4\,Tr(A)$

$\boxed{Tr(\alpha A) = \alpha\, Tr(A)}$

$\alpha \to$ constant

Question:

The trace of matrix A is 3 and the trace of matrix B is -4. Find the trace of matrix $A + B$

$3 + (-4) = \boxed{-1}$

↳ $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$; $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$

$Tr(A) = a_{11} + a_{22}$ $Tr(B) = b_{11} + b_{22}$

$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$

$Tr(A + B) = a_{11} + b_{11} + a_{22} + b_{22} = Tr(A) + Tr(B)$

Question:

The trace of matrix A is -1 and the trace of matrix B is 3. Find the trace of matrix $2A - 3B$.

↳ $Tr(A) = -1, \quad Tr(B) = 3$ $\qquad X = 2A$

$\qquad Y = 3B$

$Tr(2A - 3B) = Tr(X - Y)$

$= Tr(X) - Tr(Y)$

$= Tr(2A) - Tr(3B) = 2\,Tr(A) - 3\,Tr(B)$

$= 2(-1) - 3(3) = -11$

$$\boxed{Tr[\alpha A + \gamma B] = \alpha\, Tr(A) + \gamma\, Tr(B) \;;\; \alpha \text{ and } \gamma \text{ are scalars.}}$$

↳ Multiplication of two Matrices:-

Given:- $A = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}_{3\times 3}$; $B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & -2 \end{bmatrix}_{3\times 3}$

Find $Y = AB$ $\quad Z = BA$

↳

$$Y = AB = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 7 & 3 & 5 \\ 2 & -2 & 2 \\ 2 & 5 & 0 \end{bmatrix}_{3\times 3}$$

$$= \begin{bmatrix} (3\times 1)+(2\times 2)+(-1\times 0) & (3\times 2)+[2\times(-1)]+[(-1)\times 1] & (3\times 1)+(2\times 0)+(-1)(-2) \\ (0\times 1)+(1\times 2)+(-1\times 0) & (0\times 2)+(1\times -1)+(-1\times 1) & (0\times 1)+(1\times 0)+(-1\times -2) \\ (2\times 1)+(0\times 2)+(1\times 0) & (2\times 2)+(0\times -1)+(1\times 1) & (2\times 1)+(0\times 0)+(1\times -2) \end{bmatrix}$$

$$2 = BA = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\boxed{A+B = B+A}$$

$$\boxed{AB \neq BA}$$

$$BA = \begin{bmatrix} 5 & 4 & -2 \\ 6 & 3 & -1 \\ -4 & 1 & -3 \end{bmatrix}_{3\times 3}$$

↳ $\text{Trace}(AB) = 7 - 2 + 0 = 5$

$\text{Trace}(BA) = 5 + 3 - 3 = 5$

↳ $\boxed{\text{Trace}(AB) \neq Tr(A) \cdot Tr(B)}$

$$\boxed{\begin{gathered} Tr(AB) = Tr(BA) \\ Tr(\alpha A + \gamma B) = \alpha Tr(A) + \gamma Tr(B) \end{gathered}}$$

$Tr(A) = 5, \quad Tr(B) = -2$

Given :- $A = \begin{bmatrix} 3 & 2 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 2 & 0 & 1 & 2 \end{bmatrix}_{3\times4}$; $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}_{4\times2}$

find $Y = AB$ $\quad Z = BA$

↳

$$AB = \begin{bmatrix} 3 & 2 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 2 & 0 & 1 & 2 \end{bmatrix}_{3\times4} \begin{bmatrix} 1 & 2 \\ 2 & -1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}_{4\times2} = \begin{bmatrix} 7 & 3 \\ 3 & -3 \\ 4 & 3 \end{bmatrix}_{3\times2}$$

$$Y = AB = (3\times4)(4\times2)$$
$$Y = (3\times2)$$

↳ $[A]_{m\times n} \; [B]_{n\times p}$

$[AB]_{m\times p}$

$$Z = BA = \begin{bmatrix} 1 & 2 \\ 2 & -1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}_{4\times 2} \begin{bmatrix} 3 & 2 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 2 & 0 & 1 & 2 \end{bmatrix}_{3\times 4} = \begin{bmatrix} (1\times 3)+(2\times 0)+(2\times 2) \\ \\ \end{bmatrix}$$

$\downarrow$

$\boxed{\text{NOT Possible}}$

$$[B]_{4\times \underline{2}}\ [A]_{\underline{3}\times 4} \rightarrow \boxed{\text{NO MULTIPLICATION POSSIBLE}}$$

NOTE:- Let order of Matrix $P = m\times n$

Let " " " $Q = n\times p$

Let " " " $R = p\times m$

then $PQ : (m\times \underline{n})(\underline{n}\times p) \rightarrow (m\times p)$; $QP : (n\times \underline{p})(\underline{m}\times n) \rightarrow \times$

$QR : (n\times \underline{p})(\underline{p}\times m) \rightarrow (n\times m)$; $RQ : (p\times \underline{m})(\underline{n}\times p) \rightarrow \times$

$PR : (m\times \underline{n})(\underline{p}\times m) \rightarrow \times$; $RP : (p\times \underline{m})(\underline{m}\times n) \rightarrow (p\times n)$

↳ $PQR : (m\times n)(n\times p)(p\times m) : (m\times m)$

↳ $PRQ : (m\times n)(p\times m)(n\times p) \to \times$

↳ $RPQ : (p\times m)(m\times n)(n\times p) \to (p\times p)$

↳ $RQP : (p\times m)(n\times p)(m\times n) \to \times$

↳ $QPR : (n\times p)(m\times n)(p\times m) \to \times$

↳ $QRP : (n\times p)(p\times m)(m\times n) \to (n\times n)$

* **Properties** :-

1. $AB \neq BA$

2. $ABC = (AB)C = A(BC)$

3. $(A+B)C = AC + BC$

4. $A^2 = A \cdot A$

$$\boxed{\begin{array}{ll} Y = ABC = (AB)C = A(BC) & \\ [A]\,\underbrace{[B]\,[C]} & \underbrace{[A]\,[B]}\,[C] \\ \underbrace{[A]}\,\underbrace{[X]} & [X]\,[C] \\ = Y & = Y \end{array}}$$

Q. $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 0 & -2 & 1 \end{bmatrix}$; Find $A^2 = ?$

↳ $A^2 = \begin{bmatrix} 4 & 1 & 9 \\ 1 & 1 & 1 \\ 0 & 4 & 1 \end{bmatrix} \rightarrow \times$

wrong

$$A^2 = A \cdot A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -5 & 10 \\ 1 & 0 & - \ \ - \\ - & - & - \end{bmatrix}_{3 \times 3}$$

Q. Correct statement?

✓ (a) $A^3 = A^2 \cdot A$

✓ (b) $A^4 = A^3 \cdot A = A^2 \cdot A^2 = A \cdot A^3$

✗ (c) $(AB)^2 = A^2 B^2$

✓ (d) $(AB)^2 = A(BA)B$

✓ (e) Given that $AB = BA$; then $(AB)^2 = A^2 B^2$

(a) $A^3 = \underbrace{A \cdot A} \cdot A = A^2 \cdot A$

(b) $A^4 = A \cdot A \cdot A \cdot A = A^3 \cdot A = A^2 \cdot A^2 = A \cdot A^3$

(c) $(AB)^2 = (AB)(AB) = ABAB = A(BA)B$

if $AB = BA \Rightarrow (AB)^2 = A(AB)B = AABB = A^2 B^2$

* Multiplying any Matrix with identity (square) Matrix :-

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 0 \\ 0 & -2 \end{bmatrix}_{3\times 2} \quad ; \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}$$

↳ $AI_2 = \begin{bmatrix} 3 & -1 \\ 1 & 0 \\ 0 & -2 \end{bmatrix}_{3\times 2} = A$

$$\boxed{AI_2 = A}$$

** $\boxed{AI = IA = A}$

↳ $IA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3} \begin{bmatrix} 3 & -1 \\ 1 & 0 \\ 0 & -2 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 3 & -1 \\ 1 & 0 \\ 0 & -2 \end{bmatrix} = A$

↳ **Multiplication of **diagonal Matrices:-**

$$A = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}_{2\times 2} \quad ; \quad B = \begin{bmatrix} \gamma & 0 \\ 0 & \eta \end{bmatrix}_{2\times 2}$$

$A\ B$

$(2\times 2)\ (2\times 2) : (2\times 2)$

$B\ A$

$(2\times 2)\ (2\times 2) : (2\times 2)$

$$AB = \begin{bmatrix} \alpha\gamma & 0 \\ 0 & \beta\eta \end{bmatrix} \qquad BA = \begin{bmatrix} \gamma\alpha & 0 \\ 0 & \eta\beta \end{bmatrix}$$

$$\boxed{I_n \cdot I_n \cdot I_n \cdots = I_n}$$

$\rightsquigarrow$ Diagonal

Q. $A = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$; Find Trace $[A^7]$

↳ $A^7 = A \cdot A \cdot A \cdot A \cdot A \cdot A \cdot A = \begin{bmatrix} (-3)^7 & 0 & 0 \\ 0 & (-2)^7 & 0 \\ 0 & 0 & (2)^7 \end{bmatrix} = A^7$

$Tr[A^7] = (-3)^7 + (-2)^7 + (2)^7$

$= -(3)^7$

= Ans

★ $Tr[\alpha A + \gamma B] = \alpha\, Tr(A) + \gamma\, Tr(B)$

★ $Tr[AB] = Tr[BA]$

Q. Matrix A has $Tr(A) = 3$; $Tr(AB) = ? \neq Tr(A) \cdot Tr(B)$

Matrix B has $Tr(B) = 2$ $\quad = 3 \times 2 = 6$ ✗

↓

can't be determined.

Q. The order of matrix A is 3×4 and the order of matrix B is 4×5. Find the number of multiplications and additions required to compute the product AB.

↳ $[A]_{3\times4} \quad [B]_{4\times5}$

\# multiplications $= 3 \times 4 \times 5 = 60$

\# Addition $= 3 \times (4-1) \times 5 = 45$

$$\boxed{\begin{array}{l} [A]_{m\times n} \quad [B]_{n\times p} \\ \text{\# Multiplications} = m \times n \times p \\ \text{\# Additions} = m \times (n-1) \times p \end{array}}$$

↳ Transpose of a Matrix :-

Rows ↔ Column , Column ↔ Rows

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}_{3\times 3} ; \quad A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3\times 3}$$

$A : (m \times n)$

$A^T : (n \times m)$

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{3\times 2} ; \quad A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times 3}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}_{1\times 3} ; \quad A^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3\times 1}$$

N.B.-

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$Y = AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \Rightarrow \underset{Y^T}{(AB)^T} = \begin{bmatrix} 19 & 43 \\ 22 & 50 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$B^T . A^T = \begin{bmatrix} 19 & 43 \\ 22 & 50 \end{bmatrix}$$

$$\boxed{(AB)^T = B^T . A^T}$$

$$(ABC)^T = (YC)^T = C^T Y^T = C^T B^T A^T$$

$$Y = AB \; ; \quad Y^T = (AB)^T = B^T A^T$$

$$\boxed{(ABC)^T = C^T B^T A^T}$$

$$\boxed{(ABCD \dots Z)^T = Z^T \dots D^T C^T B^T A^T}$$

↳ Properties :-

1. $(A^T)^T = A$

2. $(A+B)^T = A^T + B^T$

3. $(KA)^T = K \cdot A^T$; K : Scalar

4. $(AB)^T = B^T \cdot A^T$

5. $(ABCD)^T = D^T C^T B^T A^T$

* Some Special Matrices:-

1. Orthogonal Matrix:-

↳ Matrix A is said to be orthogonal if

$$\boxed{AA^T = I = A^T A}$$

Eg.

$$A = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

$$A^T = \begin{bmatrix} \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A^T A$$

- Important observation:-

$$A = \begin{bmatrix} 2/3 & 1/3 & 2/3 \\ -2/3 & 2/3 & 1/3 \\ 1/3 & 2/3 & -2/3 \end{bmatrix}$$

$$\boxed{\begin{gathered} R_1R_2 = 0 = R_2R_3 = R_1R_3 \\ C_1C_2 = 0 = C_2C_3 = C_1C_3 \end{gathered}}$$

$$\boxed{\Sigma R_1^2 = \Sigma R_2^2 = \Sigma R_3^2 = \Sigma C_1^2 = \Sigma C_2^2 = \Sigma C_3^2 = 1}$$

If A is orthogonal:-

1. $\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = 1 = \left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2$

$= \left(\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2$

2. $\frac{2}{3}\left(-\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{2}{3}\right)\left(\frac{1}{3}\right) = 0 = \left(-\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)$

$= \left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{2}{3}\right)\left(-\frac{2}{3}\right) = \left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(-\frac{2}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = 0 = \cdots$

For any orthogonal metrix of order $m \times n$:-

$$A = \begin{bmatrix} a_{11} & a_{12} & - - - - & a_{1n} \\ a_{21} & a_{22} & - - - - - & a_{2n} \\ a_{31} & a_{32} & - - - \cdot - - & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & - - - - - & a_{mn} \end{bmatrix}_{m \times n}$$

↳ $a_{11}^2 + a_{12}^2 + --- a_{1n}^2 = 1 = a_{21}^2 + a_{22}^2 + --- a_{2n}^2 = ---$

↳ $a_{11}^2 + a_{21}^2 + --- a_{m1}^2 = 1 = a_{12}^2 + a_{22}^2 + -- a_{m2}^2 = ----$

↳ $a_{11}a_{12} + a_{21}a_{22} + --- a_{m1}a_{m2} = 0 = a_{11}a_{21} + a_{12}a_{22} + --- a_{1n}a_{2n} =$ ---

Q.

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & a & 0 \\ \frac{1}{\sqrt{2}} & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

is an orthogonal matrix.

Find the values of a, b, c.

(A) $a = \frac{1}{\sqrt{2}},\ b = -\frac{1}{\sqrt{2}},\ c = 1$ or -1

(B) $a = -\frac{1}{\sqrt{2}},\ b = \frac{1}{\sqrt{2}},\ c = 1$ or -1

(C) $a = \frac{1}{\sqrt{2}},\ b = \frac{1}{\sqrt{2}},\ c = -1$

(D) $a = -\frac{1}{\sqrt{2}},\ b = -\frac{1}{\sqrt{2}},\ c = 1$

$\frac{1}{\sqrt{2}}(a) + \frac{1}{\sqrt{2}}(b) + 0(0) = 0$

$\boxed{a = -b}$ ✓

$\frac{1}{2} + ab + 0 = 0$

$\boxed{ab = -\frac{1}{2}}$

$\hookrightarrow a^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 0^2 = 1$

$a^2 + \frac{1}{2} = 1$

$\boxed{a = \pm\frac{1}{\sqrt{2}}}$

$\boxed{b = \pm\frac{1}{\sqrt{2}}}$

$c^2 = 1 \Rightarrow \boxed{c = \pm 1}$ ✓

$a = \frac{1}{\sqrt{2}},\ b = -\frac{1}{\sqrt{2}},\ c = 1$

$a = \frac{1}{\sqrt{2}},\ b = -\frac{1}{\sqrt{2}},\ c = -1$

$a = -\frac{1}{\sqrt{2}},\ b = \frac{1}{\sqrt{2}};\ c = \pm 1$

N.B. :- 1. Matrix A is said to be orthogonal if $\boxed{AA^T = A^T A = I}$

2. $[A]_{m\times n}$ and $[B]_{m\times p}$ are said to be orthogonal if

$$\boxed{A^T B = B^T A = [0]} \neq AB^T \neq BA^T$$

$A^T B = [0]$

$[A^T B]^T = [0]^T$

$B^T A = [0]$

Q. $A = \begin{bmatrix} 1 \\ 2 \\ \alpha \end{bmatrix}$; $B = \begin{bmatrix} -1 \\ \alpha \\ 3 \end{bmatrix}$; Given that A and B are orthogonal to each other ; find α ?

v_1 v_2

↳ $A^T B = B^T A = [0]$

↳ $A^T B = [1 \quad 2 \quad \alpha]_{1\times 3} \begin{bmatrix} -1 \\ \alpha \\ 3 \end{bmatrix}_{3\times 1} = [-1 + 2\alpha + 3\alpha]_{1\times 1}$

$= [-1 + 5\alpha]_{1\times 1} = [0]$

↳ $5\alpha - 1 = 0$

$\boxed{\alpha = 0.2}$

2. Symmetric Matrix :-

Matrix A is known to be symmetric if $A^T = A$

$$A = \begin{bmatrix} a & e & f \\ e & b & g \\ f & g & c \end{bmatrix}_{3\times3}$$

3. Skew Symmetric Matrix :- $A^T = -A$, $-A^T = A$

$$A = \begin{bmatrix} 0 & e & f \\ -e & 0 & g \\ -f & -g & 0 \end{bmatrix}_{3\times3}$$

N.B. :- A skew symmetric Matrix must have all 0's in diagonal.

Q. Which of the following statements are true?

(A) For any matrix A, AA^T is always a symmetric matrix

(B) For any matrix A, $A + A^T$ is always a symmetric matrix

(C) For any matrix A, $A - A^T$ is always a skew-symmetric matrix

(D) If A and B are symmetric, then $A + B$ and $A - B$ are always symmetric

(E) If A and B are symmetric, then AB and BA may or may not be symmetric

(F) If A and B are symmetric, then AB and BA will also be symmetric if and only if $AB = BA$

(G) If A and B are skew-symmetric, then AB and BA will also be symmetric if and only if $AB = BA$

↳ (A) $AA^T = B$ Symmetric Matrix

Symmetric Matrix : $B^T = B$ — ①

$B^T = (AA^T)^T = (A^T)^T A^T = AA^T = B$

$B^T = B$

$(AA^T)^T = AA^T$

(B) $B = A + A^T$

$B^T = (A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T = B$

$$\boxed{B^T = B}$$

(C) $B = A - A^T$

$B^T = (A - A^T)^T = A^T - (A^T)^T = A^T - A = -[A - A^T] = -B$

$\boxed{B^T = -B}$: skew symmetric

(D) Given: $A^T = A$, $B^T = B$

$Y = A + B$

$Y^T = (A + B)^T = A^T + B^T = A + B = Y$

$\Rightarrow$ $\boxed{Y^T = Y}$ → symm.

$Z = A - B$

$Z^T = A^T - B^T = A - B = Z$

$\boxed{Z^T = Z}$

(E) (F) Given: $A^T = A$; $B^T = B$

$Y = AB$

$Y^T = (AB)^T = B^T \cdot A^T = BA$

$\Rightarrow$ $\boxed{Y = AB \;,\; Y^T = BA}$

if $\boxed{AB = BA \text{ then } Y^T = Y}$ → symm. → $\boxed{AB = BA \;;\; \text{then } Z^T = Z}$

$AB \neq BA$ then $Y^T \neq Y$

$Z = BA$

$Z^T = A^T B^T = AB$

$Z = BA, \quad Z^T = AB$

$AB \neq BA$, then $Z^T \neq Z$

(G) Given $A^T = -A$; $B^T = -B$

$Y = AB$

$Y^T = B^T A^T$

$= (-B)(-A) = BA$

$Y = AB$

$Y^T = BA$

$\boxed{\text{if } AB = BA \text{ then } Y^T = Y}$

↳ symm.

Q Which of the following statements are true?

(A) If A is symmetric, then A^m is also symmetric, given m is even

(B) If A is symmetric, then A^m is also symmetric, given m is odd

(C) If A is skew-symmetric, then A^m is also skew-symmetric, given m is odd

(D) If A is skew-symmetric, then A^m is symmetric, given m is even

(E) If A and B are symmetric matrices, then $A^T BA$ is symmetric

(F) If A and B are skew-symmetric matrices, then $AB^T A$ is skew-symmetric

(G) If A and B are symmetric matrices, then $A^2 + B^2$ and $A^2 - B^2$ are symmetric

(H) If A and B are skew-symmetric matrices, then $A^2 - B^2$ is symmetric

Always True

$$\boxed{(A^m)^T = (A^T)^m}$$

↳ (A) Given $A^T = A$ — ①

$B = A^m = A \cdot A \cdot A \cdot A \cdot \text{---}$ m times

$B^T = (A^m)^T = (A \cdot A \cdot A \cdot A \cdot \text{----})^T = \text{---} A^T A^T A^T A^T = (A^T)^m$

$= \text{----} A A A A$

$= A^m = B$

$\boxed{B^T = B}$: Symm

(c) (D) $A^T = -A$: Given

$(-A)^m = -(A)^m$; $m = \text{odd}$

$B = A^m$

$(-A)^m = A^m$; $m = \text{Even}$

$B^T = (A^m)^T = (A^T)^m = (-A)^m$

↳ $B = A^m$; $B^T = (-A)^m = -A^m = -B$; $m = \text{odd} \rightarrow B^T = -B$: skew sym

$B^T = (-A)^m = A^m = B$; $m = \text{Even} \rightarrow B^T = B$: Sym.

(E) Given : $A^T = A$, $B^T = B$

$Y = A^T B A$

$Y^T = (A^T B A)^T = A^T B^T A = A^T B A = Y$

$\boxed{Y^T = Y}$ → Sym.

(F) Given $A^T = -A$, $B^T = -B$

$Z = A B^T A$

$Z^T = (A B^T A)^T = A^T B A^T$

$= (-A)(-B)^T(-A)$

$= -A B^T A = -Z$

skew sym ← $\boxed{Z^T = -Z}$

(G) $A^T = A$, $B^T = B$

$Y = A^2 + B^2$

$Y^T = (A^2 + B^2)^T$

$= (A^2)^T + (B^2)^T$

$= (A^T)^2 + (B^T)^2$

$\boxed{Y^T = A^2 + B^2 = Y}$ → symm ← $\boxed{Z^T = A^2 - B^2 = Z}$

$Z = A^2 - B^2$

$Z^T = (A^2 - B^2)^T$

$= (A^2)^T - (B^2)^T$

$= (A^T)^2 - (B^T)^2$

(H) $A^T = -A$, $B^T = -B$

$Y = A^2 - B^2$

$Y^T = (A^2)^T - (B^2)^T$

$Y^T = (A^T)^2 - (B^T)^2 = (-A)^2 - (-B)^2 = A^2 - B^2 = Y$

$\boxed{Y^T = Y}$

↳ symmetric

Question

Which of the following statements are true?

(A) For a symmetric matrix of order n, the maximum number of unique elements is $\frac{n(n+1)}{2}$

(B) For a skew-symmetric matrix of order n, the maximum number of unique elements is $\frac{n(n-1)}{2}$

(C) For a symmetric matrix of order n, the maximum number of unique elements is $\binom{n+1}{2}$, and for a skew-symmetric matrix, it is $\binom{n}{2}$

(A)

$$\begin{bmatrix} a & e & f \\ e & b & g \\ f & g & c \end{bmatrix}_{3\times 3} \qquad 3+1+2$$

$$\begin{bmatrix} a & e & f & h \\ e & b & g & i \\ f & g & c & k \\ h & i & k & d \end{bmatrix}_{4\times 4} \qquad 4+1+2+3$$

$$\begin{bmatrix} a & f & g & i & l \\ & b & h & j & m \\ & & c & k & n \\ & & & d & o \\ & & & & e \end{bmatrix}_{5\times 5} \qquad 5+1+2+3+4$$

n order: $n+1+2+3+ \ldots n-1 = 1+2+3+ \ldots n$

$$= \frac{n(n+1)}{2} = {}^{n+1}C_2 = \binom{n+1}{2}$$

↳ For skew symm., diagonal elements = 0

n order skew symm. matrix ⇒ $1+2+3+ \dots (n-1)$

$$\frac{(n-1)(n)}{2} = \frac{n(n-1)}{2} = {}^{n}C_2 = \binom{n}{2}$$

<u>N.B.</u> - writing matrix A into its symmetric and skew-symmetric parts.

$$\boxed{A = \underbrace{\left(\frac{A + A^T}{2}\right)}_{\text{Symm.}} + \underbrace{\left(\frac{A - A^T}{2}\right)}_{\text{Skew-Symm.}}}$$

* Taking conjugate of a Matrix:- $j \rightarrow -j$; $-j \rightarrow j$

$$A = \begin{bmatrix} 2+j & 3 & 3+\sqrt{2} \\ 3j & -2+j & -1 \end{bmatrix} ; \quad A^* = \begin{bmatrix} 2-j & 3 & 3+\sqrt{2} \\ -3j & -2-j & -1 \end{bmatrix}$$

* Transposed Conjugate / Tranjugated of Matrix A:-

$$A^\theta = (A^*)^T = (A^T)^*$$

$$A^\theta = (A^*)^T = \begin{bmatrix} 2-j & -3j \\ 3 & -2-j \\ 3+\sqrt{2} & -1 \end{bmatrix} ; \quad (A^T)^* = \begin{bmatrix} 2-j & -3j \\ 3 & -2-j \\ 3+\sqrt{2} & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2+j & 3 & 3+\sqrt{2} \\ 3j & -2+j & -1 \end{bmatrix} \quad ; \quad A^* = \begin{bmatrix} 2-j & 3 & 3+\sqrt{2} \\ -3j & -2-j & -1 \end{bmatrix}$$

$$A^{\theta} = (A^*)^T = (A^T)^* = \begin{bmatrix} 2-j & -3j \\ 3 & -2-j \\ 3+\sqrt{2} & -1 \end{bmatrix}$$

Tranjugate

$$(KA)^T = \underbrace{K}_{\text{scalar}} \cdot A^T$$

Properties:-

1. $(A^{\theta})^{\theta} = A$
2. $(A+B)^{\theta} = A^{\theta} + B^{\theta}$
3. $(ABCD)^{\theta} = D^{\theta} C^{\theta} B^{\theta} A^{\theta}$
4. $(KA)^{\theta} = K^* A^{\theta}$

↳ Let $A = \begin{bmatrix} 2+j & -1 \\ -j & 3+j \end{bmatrix}$; Let $K = 1+j$; $K^* = 1-j$

↳

$$B = KA = \begin{bmatrix} (2+j)(1+j) & -(1+j) \\ -j(1+j) & (3+j)(1+j) \end{bmatrix}$$

$$B^\theta = (B^*)^T = \begin{bmatrix} (2-j)(1-j) & j(1-j) \\ -(1-j) & (3-j)(1-j) \end{bmatrix} = (1-j)\begin{bmatrix} 2-j & j \\ -1 & 3-j \end{bmatrix}$$

$$= (1-j)\, A^\theta$$

↳ $A^\theta = (A^*)^T = \begin{bmatrix} 2-j & j \\ -1 & 3-j \end{bmatrix}$

$$= K^* A^\theta$$

$$B^\theta = K^* A^\theta$$

$$\boxed{(KA)^\theta = K^* A^\theta}$$

Special Type of Matrices :-

1. Unitary Matrix :-

A square Matrix A is said to be unitary if $A \cdot A^{\theta} = I = A^{\theta} \cdot A$

Eg.-

$$A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}_{2\times 2} \qquad A^{\theta} = \begin{bmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ \frac{-1-i}{2} & \frac{1+i}{2} \end{bmatrix} = (A^{*})^{T}$$

$$AA^{\theta} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

* Important observation:-

$(a+b)^* = a^* + b^*$

$(ab)^* = a^* b^*$

Let Matrix $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ is unitary

$(a^*)^* = a$

then:- ↳ $|a|^2 + |b|^2 + |c|^2 = 1 = |d|^2 + |e|^2 + |f|^2 = |g|^2 + |h|^2 + |i|^2 = |a|^2 + |d|^2 + |g|^2$

$= |b|^2 + |e|^2 + |h|^2 = |c|^2 + |f|^2 + |i|^2$

↳ $ab^* + de^* + gh^* = 0 = bc^* + ef^* + hi^* = ac^* + df^* + gi^*$

$= a^*b + d^*e + g^*h = 0 = b^*c + e^*f + h^*i = a^*c + d^*f + g^*i$

$= a^*d + b^*e + c^*f = 0 = d^*e + e^*h + f^*i = - - - - -$

2. Hermition Matrix :- $A^{\theta} = A$

Symm. → $A^T = A$

$$A = \begin{bmatrix} \underline{a} & d & e \\ g & \underline{b} & f \\ h & i & \underline{c} \end{bmatrix}$$

$$A^{\theta} = \begin{bmatrix} \underline{a}^* & g^* & h^* \\ d^* & \underline{b}^* & i^* \\ e^* & f^* & \underline{c}^* \end{bmatrix}$$

$$\boxed{\begin{aligned} a &= a^* \\ b &= b^* \\ c &= c^* \end{aligned}}$$

$d = g^*$

$d^* = g$

$$A = \begin{bmatrix} \underline{a} & d & e \\ d^* & \underline{b} & f \\ e^* & f^* & \underline{c} \end{bmatrix}$$

↳ $\boxed{\text{Hermition Matrix}}$

↳ $\boxed{\text{Diagonal elements has to be real}}$

3. Skew-Hermition Matrix :- $A^{\theta} = -A$ or $-A^{\theta} = A$

skew sym. $\rightarrow A^T = -A$

$$A = \begin{bmatrix} a & d & e \\ g & b & f \\ h & i & c \end{bmatrix} \qquad -A^{\theta} = \begin{bmatrix} -a^* & -g^* & -h^* \\ -d^* & -b^* & -i^* \\ -e^* & -f^* & -c^* \end{bmatrix}$$

$$\boxed{\begin{aligned} a &= -a^* \\ b &= -b^* \\ c &= -c^* \end{aligned}}$$

$$A = \begin{bmatrix} a & d & e \\ -d^* & b & f \\ -e^* & -f^* & c \end{bmatrix}$$

a, b, c :- purely Img. or zero

$\hookrightarrow$ $\boxed{\text{Diagonal elements has to be purely Img. or zero.}}$

Q. Given $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}_{2\times 2}$; $B = e^A$ $\qquad [B] = e^{[A]}$

Find the summation of all the elements of Matrix B.

↳ $[B] = ?$ $\qquad [B] = e^{[A]}$

$$= \begin{bmatrix} e^0 & e^1 \\ e^0 & e^0 \end{bmatrix} = \begin{bmatrix} 1 & e \\ 1 & 1 \end{bmatrix} \times \times$$

wrong

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$e^{[A]} = I + [A] + \frac{[A]^2}{2!} + \frac{[A]^3}{3!} + \frac{[A]^4}{4!} + \cdots$$

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; $[A] = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\qquad [A]^2 = [A][A] = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$I \checkmark$, $A \checkmark$, $A^2 = 0$, $A^3 = A \cdot A^2 = A \cdot [0] = [0]$

$A^4 = A \cdot A^3 = A \cdot [0] = [0] = \dots = A^5$

$$e^{[A]} = I + [A]$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

summation of all elements $= 1 + 1 + 1 + 0 = 3 = \underline{\text{ANS}}$

Q.

Question

Which of the following statements are true for a complex square matrix A?

Option A:
A is skew-Hermitian **iff** iA is Hermitian

Option B:
A is Hermitian **iff** iA is skew-Hermitian

Option C:
If A is skew-Hermitian, then e^A is unitary

Option D:
If A is Hermitian, then e^A is unitary

Iff = If and only If

x is True iff y is True

⇕

If x is True then y is True

and

If y is True then x is True

↳ (A) (1) If A is skew Hermition then iA is Hermition.

Given; $A^{\theta} = -A$, $B = iA$

$B^{\theta} = (iA)^{\theta}$

$B^{\theta} = \bar{i} A^{\theta} = -iA^{\theta}$

$B^{\theta} = -i(-A) = iA = B$

$\boxed{B^{\theta} = B}$

$(iA)^{\theta} = iA$ ✓

(2) If iA is Hermition then A is skew Hermition.

Given $(iA)^\theta = iA$

$-\cancel{i} A^\theta = \cancel{i} A$

$\boxed{A^\theta = -A}$ → A is skew Hermition

(B.) Homework

(C) Given $A^\theta = -A$

$e^A = ?$: $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \cdots = B$

$B^\theta = (e^A)^\theta = I^\theta + A^\theta + \frac{(A^2)^\theta}{2!} + \frac{(A^3)^\theta}{3!} + \frac{(A^4)^\theta}{4!} + \cdots$

B is unitary
$BB^\theta = I$

$(A+B)^\theta = A^\theta + B^\theta$

$(A^\theta)^n = (A^n)^\theta$

$$B^{\dagger} = \underbrace{I + (-A) + \frac{(-A)^2}{2!} + \frac{(-A)^3}{3!} + \frac{(-A)^4}{4!} + \cdots}_{} = e^{-A}$$

$$B^{\dagger} = (e^{A})^{\dagger} = e^{-A}$$

$$B \cdot B^{\dagger} = e^{A} \cdot e^{-A} = e^{A-A} = e^{0} = I$$

$\boxed{B \cdot B^{\dagger} = I}$ $B = e^{A}$ is unitary

(b) Given $A^{\dagger} = A$; $B = e^{A}$

$$e^{A} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \quad ; \quad (e^{A})^{\dagger} = I + A^{\dagger} + \frac{(A^{\dagger})^2}{2!} + \frac{(A^{\dagger})^3}{3!} + \cdots$$

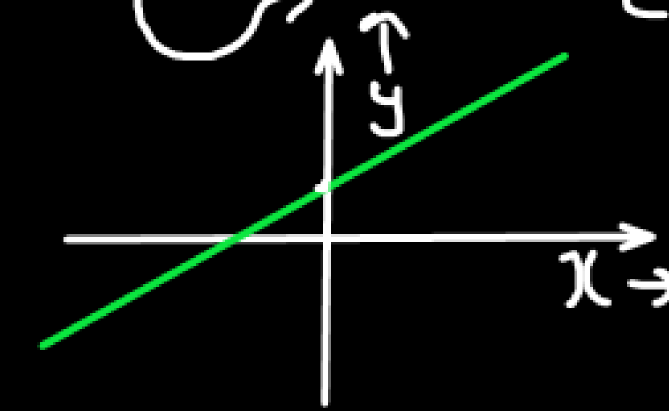
$$= I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

$$(e^A)^{\dagger} = e^A$$

$$\boxed{(e^A)(e^A)^{\dagger} = e^A \cdot e^A = e^{2A} \neq I}$$ $\rightsquigarrow$ e^A is not unitary

* Understanding the meaning of linearity :-

$y^1 = 2x^1 + 4$ linear eqn



↳ $A = [2 \quad 3 \quad 7]$

$B = [4 \quad 6 \quad 14]$

→ $B = 2A$; $A = \frac{1}{2}B$

* B is linearly depending on A
* A is linearly depending on B
* A and B are linearly dependent on each other.

↳

$A = [1 \quad 4 \quad 7]$

$B = [2 \quad 8 \quad 11]$

$\alpha A = [\alpha \quad 4\alpha \quad 7\alpha] \neq [2 \quad 8 \quad 11]$

$\boxed{[B] \neq \alpha [A]}$ $\boxed{[A] \neq \gamma [B]}$

* A and B are independent of each other.
* A and B are two linearly independent Matrices.

$A = [1 \quad 4 \quad 7]$

$B = [2 \quad -1 \quad 1]$

$C = [5 \quad 2 \quad 9]$

$\alpha[A] + \beta[B] = [C]$

Independent

dependent

$C = A + 2B$

OR

$A = C - 2B$

OR

$B = 0.5C - 0.5A$

$\alpha[C] \neq A$

$\alpha[C] \neq B$

$\alpha[B] \neq A$

$\alpha[B] \neq C$

$\alpha[A] \neq B$

$\alpha[A] \neq C$

no. of independent Matrices = 2

" " dependent " = 1

* C is linearly depending on A and B.
* B is linearly depending on A and C.
* A is linearly depending on A and B.

↳ There are two linearly independent Matrix and one linearly dependent Matrix.

↳ $A = [2 \quad 3 \quad -1]$

$B = [1 \quad 1.5 \quad -0.5]$

$C = [3 \quad 4.5 \quad -1.5]$

Independent

$A/2 = B$

$3/2 A = C$

dependent

$2B = A$

$3B = C$

$2/3 C = A$

$1/3 C = B$

↳ No. of Lineary Independent Matrices = 1

↳ No. of " dependent " = 2

* A is linearly depending on B
* A is linearly depending on C
* B is linearly depending on A
* B is linearly depending on C
* C is linearly depending on A
* C is linearly depending on B

↳ There is one linearly independent vector (Matrix) and two linearly dependent vector (Matrices).

↳ A linear Relation b/w P, Q and R

↳ $P = \alpha Q + \beta R$; ↳ $Q = \gamma P + \eta R$; ↳ $R = aP + bQ$

* Determinants:- ↪ Square Matrices

A **determinant** is a **single numerical value** associated with a **square matrix**.

If A is a square matrix, then its determinant is written as:

$$\det(A) \quad \text{or} \quad |A|$$

2x2 Matrix:-

Matrix ↪ $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\text{Det}(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$A = \begin{bmatrix} 2 & -1 \\ -3 & 1 \end{bmatrix} \rightsquigarrow |A| = -1$

$$|A| = \begin{vmatrix} 2 & -1 \\ -3 & 1 \end{vmatrix}$$

$$= 2(1) - (-1)(-3)$$

$$= 2 - 3 = -1$$

Finding Minors of an element:-

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad ; \det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = ?$$

Minor of a_{11}:-

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{23}a_{32}$$

Minor of a_{13}:-

$$M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = a_{21}a_{32} - a_{31}a_{22}$$

Minor of a_{22} :-

$$M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} = a_{11}a_{33} - a_{13}a_{31}$$

Minor of a_{32} :-

$$M_{32} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = a_{11}a_{23} - a_{21}a_{13}$$

Q. $A = \begin{bmatrix} 2 & 5 & 0 \\ -1 & 1 & 3 \\ -3 & -2 & 4 \end{bmatrix}$

↳ find Minor of element 3.

↳ find Minor of element 0.

↳ find Minor of element 2.

↳ Minor of element 3 = $\begin{vmatrix} 2 & 5 \\ -3 & -2 \end{vmatrix} = 2(-2) - (5)(-3)$

$= -4 + 15 = 11$

↳ Minor of element 0 = $\begin{vmatrix} -1 & 1 \\ -3 & -2 \end{vmatrix} = (-1)(-2) - (1)(-3)$

$= 2 + 3 = 5$

↳ Minor of element 2 = $\begin{vmatrix} 1 & 3 \\ -2 & 4 \end{vmatrix} = (1)(4) - (3)(-2)$

$= 4 + 6 = 10$

* Finding co-factor of an element:-

$$\text{co-factor} = \begin{cases} \text{Minor} & ; \ i+j = \text{Even} \\ -\text{Minor} & ; \ i+j = \text{odd} \end{cases}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad ; \quad \text{Det}(A) = |A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

co-factor of a_{11}:-

$$F_{11} = M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = (a_{22}a_{33} - a_{23}a_{32})$$

co-factor of a_{12}:-

$i+j = 1+2 = 3 \hookrightarrow$ odd

$$F_{12} = -M_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = -(a_{21}a_{33} - a_{31}a_{23})$$

co-factor of a_{31}:-

$$F_{31} = M_{31} = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} = (a_{12}a_{23} - a_{22}a_{13})$$

co-factor of a_{32}:-

$$F_{32} = -M_{32} = -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = -[a_{11}a_{23} - a_{21}a_{13}]$$

* Finding Determinant of Matrix A :- [order 3x3]

$$\text{Det}(A) = |A| = \begin{array}{c} \begin{array}{ccc} C_1 & C_2 & C_3 \\ \downarrow & \downarrow & \downarrow \end{array} & \\ \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} & \begin{array}{c} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{array} \end{array}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

- Finding determinant w.r.t.

1. $\underline{R_1}$:-

$$|A| = a_{11}F_{11} + a_{12}F_{12} + a_{13}F_{13}$$

$$= a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

$$= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

$$|A| = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{31}a_{23}$$
$$+ a_{13}a_{21}a_{32} - a_{13}a_{31}a_{22}$$

2. R_2 :-

3. R_3 :- ✓

$$|A| = a_{21}F_{21} + a_{22}F_{22} + a_{23}F_{23}$$
$$= -a_{21}M_{21} + a_{22}M_{22} - a_{23}M_{23}$$

$$= -a_{21}(a_{12}a_{33} - a_{32}a_{13}) + a_{22}(a_{11}a_{33} - a_{31}a_{13}) - a_{23}(a_{11}a_{32} - a_{31}a_{12})$$

4. C_1 :-

5. C_2 ✓

6. C_3 ✓

$$|A| = a_{11}F_{11} + a_{21}F_{21} + a_{31}F_{31} = a_{11}M_{11} - a_{21}M_{21} + a_{31}M_{31}$$

$$|A| = a_{11}[a_{22}a_{33} - a_{32}a_{23}] - a_{21}[a_{12}a_{33} - a_{32}a_{13}] + a_{31}[a_{12}a_{23} - a_{22}a_{13})$$

Q. a_{12}

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 1 \\ -1 & 2 & 4 \end{bmatrix}$$; Determinant of A ?

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

↳ $\underline{\underline{R_1}}$:-

$$|A| = 2[12-2] - (-1)[4+1] + 0[2+3]$$

$$= 2(10) + 1(5) = 25$$

$\underline{\underline{C_2}}$:-

$$-(-1)[4+1] + 3[8-0] - 2[2-0]$$

$$= 5 + 24 - 4 = 25$$

$\underline{\underline{C_3}}$:-

$$|A| = 0[\ \ ] - 1(4-1) + 4(6+1)$$

$$= -3 + 28 = \boxed{25}$$

Q.

$$A = \begin{bmatrix} 17 & 3 & -99 \\ 7 & 0 & 3 \\ 2 & 0 & 4 \end{bmatrix} \qquad |A| = ? \qquad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

↳ $\underline{\underline{C_2}}$:-

$$|A| = -3(28-6) + 0[\quad] - 0[\quad]$$

$$= -66$$

$\underline{\underline{R_3}}$:-

$$|A| = 2[9] - 0[\quad] + 4[0-21]$$

$$= 18 - 84 = \underline{\underline{-66}}$$

Q.26	Consider the matrix $M = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{bmatrix}$. Which of the following options is/ are TRUE if $\det(M) \neq 0$?
(A)	$a = -\frac{1}{2}$ and $b = -\frac{1}{2}$
(B)	$a = \frac{1}{2}$ and $b = \frac{1}{2}$
(C)	$a = -3$ and $b = 0$
(D)	$a = \frac{1}{2}$ and $b = -3$

$$|M| = \begin{array}{ccc} & & c_3 \\ \end{array}\begin{vmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{vmatrix} \neq 0$$

$$\begin{array}{c} c_3 \\ + \\ - \\ + \end{array}$$

$$|M| = 1[a+3] - 0[\ ] + b[5]$$

$$= a + 3 + 5b$$

$$|M| \neq 0$$

$$\underbrace{a + 3 + 5b \neq 0}$$

(a) $-\frac{1}{2} + 3 + 5\left(-\frac{1}{2}\right) = 2.5 - 2.5 = 0$

(b) $\frac{1}{2} + 3 + 5\left(\frac{1}{2}\right) = 2.5 + 3.5 = 6 \neq 0$

(c) $-3 + 3 + 5(0) = 0$

(d) $\frac{1}{2} + 3 + 5(-3) = 3.5 - 15 \neq 0$

↳ **Properties of Determinants:-**

$|A+B+C+D| \neq |A|+|B|+|C|+|D|$

1. $|A+B| \neq |A|+|B|$

$|ABCD| = |A||B||C||D|$

2. $|AB| = |A|\cdot|B|$

↳

$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ $\quad B = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$; $|A| = -7$; $|B| = 4$

$|A|+|B| = -3$

$A+B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ $\quad AB = \begin{bmatrix} 3 & 5 \\ 2 & -6 \end{bmatrix}$

$|A|\cdot|B| = -28$

$|A+B| = 4-4 = 0$ $\quad |AB| = -28 = |A||B|$

$|A+B| = 0 \neq |A|+|B|$;

3. $|I_n| = 1$

4. $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 5$

$R_1 \to 3R_1$

$B = \begin{bmatrix} 6 & 9 \\ 1 & 4 \end{bmatrix} \Rightarrow |B| = \begin{vmatrix} 6 & 9 \\ 1 & 4 \end{vmatrix} = 15 = 5 \times 3$

$R_2 \to 3R_2$

$C = \begin{bmatrix} 2 & 3 \\ 3 & 12 \end{bmatrix} \Rightarrow |C| = \begin{vmatrix} 2 & 3 \\ 3 & 12 \end{vmatrix} = 15 = 5 \times 3$

$C_1 \to 3C_1$

$D = \begin{bmatrix} 6 & 3 \\ 3 & 4 \end{bmatrix} \Rightarrow |D| = \begin{vmatrix} 6 & 3 \\ 3 & 4 \end{vmatrix} = 15 = 5 \times 3$

$C_2 \to 3C_2$

$E = \begin{bmatrix} 2 & 9 \\ 1 & 12 \end{bmatrix} \Rightarrow |E| = \begin{vmatrix} 2 & 9 \\ 1 & 12 \end{vmatrix} = 15 = 5 \times 3$

↳

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \alpha$$

$$\underset{\sim}{k} \cdot |A| = \begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ka_{31} & ka_{32} & ka_{33} \end{vmatrix}$$

$$K \cdot \alpha$$

$$= \begin{vmatrix} ka_{11} & a_{12} & a_{13} \\ ka_{21} & a_{22} & a_{23} \\ ka_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & ka_{12} & a_{13} \\ a_{21} & ka_{22} & a_{23} \\ a_{31} & ka_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & ka_{13} \\ a_{21} & a_{22} & ka_{23} \\ a_{31} & a_{32} & ka_{33} \end{vmatrix}$$

Q. $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix}$; Matrix $B = -2A$

Find $Det(B) = ?$

$Det(A) = ?$

$[B] = [-2A] = 2[A]$

$|B| = \alpha |A|$

?

$|A| = 1(3) + 1(2) = 5$ ✓

$B = -2A = \begin{bmatrix} -2 & 2 & 0 \\ -4 & -2 & 2 \\ 0 & -4 & -2 \end{bmatrix}$

$|B| = -2(4+8) - 2(8)$

$= -24 - 16 = -40$ ✓

$B = -2A$

$|B| = |-2A|$

$\begin{vmatrix} -2 & 2 & 0 \\ -4 & -2 & 2 \\ 0 & -4 & -2 \end{vmatrix} = (-2)(-2)(-2) \begin{vmatrix} 1 & -1 & 0 \\ 2 & 1 & -1 \\ 0 & 2 & 1 \end{vmatrix}$

$\boxed{|B| = -8|A|}$

↳ Conclusion:-

Let A is a square matrix of order $n \times n$; $\text{Det}(A) = |A|$

A new matrix $B = KA$; $[B] = K[A]$

then $\text{Det}(B) = |B| = |KA| = K^n |A|$

5. Find Determinant.

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{vmatrix} \quad ; \quad |B| = \begin{vmatrix} 2 & 4 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 3 \end{vmatrix} \quad ; \quad |C| = \begin{vmatrix} 1 & 0 & 0 \\ 4 & -2 & 0 \\ 3 & 1 & 1 \end{vmatrix}$$

$= 2(-3) = -6 = 2(-1)(3)$; $|B| = 2(-2)(3)$; $|C| = 1(-2)(1)$

↳ For Diagonal, Upper Triangular and Lower Triangular Matrices, the value of determinant is equal to the multiplication of their diagonal elements.

Q. A Matrix $B = 4I_3$; where I_3 : Identity Matrix of 3rd order

Find $|B| = ?$

↳

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad B = 4I_3 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4(4)(4) = 4^3 = 64$$

$B = 4I_3$; $|B| = |4I_3|$

$= (4)^3 \, |I_3| = 64(1) = 64$

6. $|A^T| = |A|$

$|ABCD| = |A|\,|B|\,|C|\,|D|$

7. $|A^n| = |A \cdot A \cdot A \cdot A \; \ldots \; n \text{ times}| = |A| \cdot |A| \cdot |A| \; \ldots \; n \text{ time} = |A|^n$

8. $|A| = \begin{vmatrix} 0 & 0 & 0 \\ \alpha & \beta & \gamma \\ \eta & a & b \end{vmatrix} = 0$

$|B| = \begin{vmatrix} 0 & \alpha & a \\ 0 & \beta & b \\ 0 & \gamma & c \end{vmatrix} = 0$

$$\left\{ \begin{array}{l} [A^n]^T = [A^T]^n \\ [A^n]^\theta = [A^\theta]^n \\ |A^n| = \{|A|\}^n \end{array} \right.$$

$|C| = \begin{vmatrix} \alpha & 0 & a \\ \beta & 0 & b \\ \gamma & 0 & c \end{vmatrix} = 0$

$|D| = \begin{vmatrix} 0 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} = 1(0-2) = -2$

↳ If any row (or column) of a Matrix A is completely zero.

Then $|A| = 0$

g.

$$|A| = \begin{vmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 0 & 1 & 3 \end{vmatrix} = 7 \xrightarrow{C_2 \leftrightarrow C_3} |C| = \begin{vmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ 0 & 3 & 1 \end{vmatrix} = -7$$

$$\Big\downarrow R_1 \leftrightarrow R_2$$

$$|B| = \begin{vmatrix} -1 & 1 & 2 \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix} = -7$$

4 If any two rows (or two columns) of a determinant are interchanged, then the value of determinant is multiplied by -1.

10.

$$|A| = \begin{vmatrix} 1 & 2 & -1 \\ 1 & -1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 1(5) - 2(5) - 1(5) = -10$$

$R_2 \to R_2 + R_1$

$$|B| = \begin{vmatrix} 1 & 2 & -1 \\ 2 & 1 & -3 \\ 2 & 3 & 1 \end{vmatrix} = 1(10) - 2(8) - 1(4)$$

$$= 10 - 16 - 4 = -10$$

$R_2 \to R_2 + R_1$

$$|C| = \begin{vmatrix} 1 & 2 & 0 \\ 1 & -1 & -4 \\ 2 & 3 & 2 \end{vmatrix} = -10 \qquad C_3 \to C_3^{1} - C_1^{1} + C_2^{1}$$

$$|D| = \begin{vmatrix} 1 & 2 & -1 \\ -3 & -6 & -7 \\ 2 & 3 & 1 \end{vmatrix} = -10 \qquad R_2 \to R_2^{1} + 2R_1^{1} - 3R_3^{1}$$

↳ $R_1 \to R_1 + \alpha R_2 + \beta R_3$

$R_2 \to R_2 + a R_1 + b R_3$

$R_3 \to R_3 + m R_1 + n R_2$

$C_1 \to C_1 + d C_2 + e C_3$

$C_2 \to C_2 + f C_1 + g C_3$

$C_3 \to C_3 + \eta C_1 + \gamma C_2$

$\to$: Replacing

$\leftrightarrow$: Interchange

↳ All acceptable but apply one at a time

Q.

$$|A| = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 1 & 2 \\ 1 & -3 & 4 \end{vmatrix}$$

Find Determinant.

↳ $\begin{vmatrix} 1 & 2 & -1 \\ 3 & 1 & 2 \\ 1 & -3 & 4 \end{vmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{vmatrix} 1 & 2 & -1 \\ 1 & -3 & 4 \\ 1 & -3 & 4 \end{vmatrix}$

$\downarrow R_3 \to R_3 - R_2$

$$\begin{vmatrix} 1 & 2 & -1 \\ 1 & -3 & 4 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$\boxed{|A| = 0}$

$\boxed{\text{If any two rows (or 2 columns) are identical then } |A| = 0}$

Q. $A = \begin{bmatrix} 2 & 1 \\ 4 & 6 \end{bmatrix}$ $\rightarrow$ $|A| = 12 - 4 = 8$

Perform operation $R_2 \rightarrow 2R_2 - 3R_1$ to get Matrix B

Find Det(A), Det(B) = ?

↳ $B: R_2 \rightarrow 2R_2 - 3R_1$

$\boxed{|B| = 2|A|}$

$B = \begin{bmatrix} 2 & 1 \\ 2 & 9 \end{bmatrix}$; $|B| = \underline{\underline{16}}$

$|B| = \begin{vmatrix} 2 & 1 \\ 2 & 9 \end{vmatrix}$

$$|A| = \begin{vmatrix} 2 & 1 \\ 4 & 6 \end{vmatrix} \begin{matrix} R_1 \\ R_2 \end{matrix}$$

$R_2 \to 2R_2 - 3R_1$

↳ $R_2 \to 2R_2$

$$|C| = \begin{vmatrix} 2 & 1 \\ 8 & 12 \end{vmatrix} = 2|A|$$

$$\boxed{|B| = 2|A|}$$

$R_2 \to R_2 - 3R_1$

$$|B| = \begin{vmatrix} 2 & 1 \\ 2 & 9 \end{vmatrix} = |C| = 2|A|$$

Q. Let Matrix [A] of order 3x3 with Det(A) = |A| = -3

Let Matrix [B] = 2[A]

$[C] = [A]^3$

$$[A] \xrightarrow{R_3 \leftrightarrow R_2} [D] \xrightarrow{C_1 \leftrightarrow C_2} [E]$$

$$[E] \xrightarrow{R_3 \to R_3 + 2R_1} [F]$$

$$[F] \xrightarrow{C_1 \to 3C_1 - 2C_2} [G]$$

Find $|B| + |C| + |D| + |E| + |F| + |G| = \boxed{-63}$

$|A| = -3$

$B = 2A$

$C = A^3$

$|D| = 3$, $|E| = -3$, $|F| = -3$

$|G| = -9$, $|C| = -27$, $|B| = -24$

$|D| = -|A| = -(-3) = 3$

$|E| = -|D| = -3$; $|G| = 3|F|$; $|C| = |A^3| = [|A|]^3 = (-3)^3 = -27$

$= -9$

$|E| = |F| = -3$

$|B| = |2A| = (2)^3|A| = -24$

N.B. -

$$[A] = \begin{bmatrix} a+j & b+k & c+l \\ d+m & e+n & f+o \\ g+p & h+q & i+r \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} + \begin{bmatrix} j & k & l \\ m & n & o \\ p & q & r \end{bmatrix}$$

$$\begin{vmatrix} a+j & b+k & c+l \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} j & k & l \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$\begin{vmatrix} a+j & b+k & c+l \\ d+m & e+n & f+o \\ g+p & h+q & i+r \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ d+m & e+n & f+o \\ g+p & h+q & i+r \end{vmatrix} + \begin{vmatrix} j & k & l \\ d+m & e+n & f+o \\ g+p & h+q & i+r \end{vmatrix}$$

$$\begin{vmatrix} d & e & f \end{vmatrix} + \begin{vmatrix} m & n & o \end{vmatrix} \qquad \begin{vmatrix} g & h & i \end{vmatrix} + \begin{vmatrix} p & q & r \end{vmatrix}$$

Q. Find Determinant.

(a) $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = ?$

Method - 1 :-

$R_1 \to R_1 + 2R_3$

$\begin{vmatrix} a+2x & b+2y & c+2z \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = 0$

Method 2:-

$$\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = \underbrace{\begin{vmatrix} a & b & c \\ a & b & c \\ x & y & z \end{vmatrix}}_{=0} + \begin{vmatrix} a & b & c \\ 2x & 2y & 2z \\ x & y & z \end{vmatrix}$$

$$\downarrow R_2 \to R_2 - 2R_3$$

$$= 0 \qquad \underbrace{\begin{vmatrix} a & b & c \\ 0 & 0 & 0 \\ x & y & z \end{vmatrix}}_{0}$$

$$\boxed{\text{If any two rows (or 2 columns) are propotional then } |A| = 0}$$

(b)

$$\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{vmatrix} 1 & a & bc \\ 0 & b-a & c(a-b) \\ 1 & c & ab \end{vmatrix}$$

$$\Big\downarrow R_3 \to R_3 - R_1$$

changed

$$\xrightarrow[\substack{R_3 \to R_3 - R_1 \\ \quad\ \ J \qquad J}]{R_2 \to R_2 - R_1} \begin{vmatrix} 1 & a & bc \\ 0 & b-a & c(a-b) \\ 0 & c-a & b(a-c) \end{vmatrix}$$

$$= (b-a)\,b(a-c) - c(a-b)(c-a) = (a-b)(b-c)(c-a)$$

↳ $R_1 \to R_1 + \alpha R_2 + \beta R_3$
$R_2 \to R_2 + \alpha R_3$ } ✓

? → NO

↳ $R_3 \to R_3 - 4R_1$
$R_2 \to R_2 - 3R_1 + 4R_3$ } X

? → YES

↳ $C_2 \to C_2 + 3C_1 - 4C_3$
$C_3 \to C_3 - 4C_1$ } ✓

? → NO

↳ $R_1 \to R_1 - 2R_2$
$C_2 \to C_2 - 3C_3$ } X

$R_1 \to R_1 - R_2 + 4R_4$
$R_2 \to R_2 - R_4$
$R_3 \to R_3 - 4R_4$ } ✓

(C)

$$|A| = \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0 \quad ; \text{ Find the value of } xyz \text{ ?}$$

Given $\underbrace{x \neq y \neq z \neq 0}$

4 $\quad x = 1, \quad y = -1, \quad z = z$

$$\begin{vmatrix} 1 & 1 & 2 \\ -1 & 1 & 0 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

$$2[-z^2 - z] + (1+z^3)(2) = 0$$

$$1 + z^3 - z^2 - z = 0$$

$$z^3 - z^2 - z + 1 = 0$$

$$z^2(z-1) - 1(z-1) = 0$$

$$(z-1)(z^2-1) = 0$$

$$(z-1)(z+1)(z-1) = 0$$

$$\boxed{z = 1, \quad 1, \quad -1}$$

$\hookrightarrow$ If $z = 1$ or -1, then $x = z$ or $y = z$ } NOT Acceptable.

$x = 1,\ y = -2,\ z = z$

$$\begin{vmatrix} 1 & 1 & 2 \\ -2 & 4 & -7 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0 \xrightarrow{R_2 \to R_2 + 2R_1} 3\begin{vmatrix} 1 & 1 & 2 \\ 0 & 2 & -1 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

$x = 1$

$y = -2$

$z = 1/2$

$$\boxed{xyz = -1}$$

$\underline{\underline{\text{Ans.}}}$

$$2 + 2z^3 + z^2 + z(-1-4) = 0$$

$$2z^3 + z^2 - 5z + 2 = 0$$

$$(z-1)(2z^2 + 3z - 2) = 0$$

$$(z-1)(2z^2 + 4z - z - 2) = 0$$

$$(z-1)\left[2z(z+2) - 1(z+2)\right] = 0$$

$$(z-1)(z+2)(2z-1) = 0$$

$z = 1 = x$ ✗

$z = -2 = y$ ✗

$z = 1/2$ ✓

Q. Find the area of Triangle whose vertices are

(3,8) (-4,2) (5,1)

$$\Delta = \frac{1}{2}\begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 3 & -4 & 5 \\ 8 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$R_2 \to R_2 - R_1$

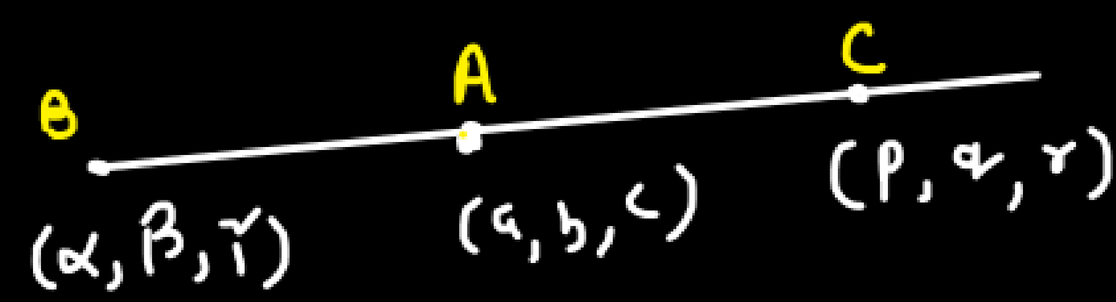
$R_3 \to R_3 - R_1$

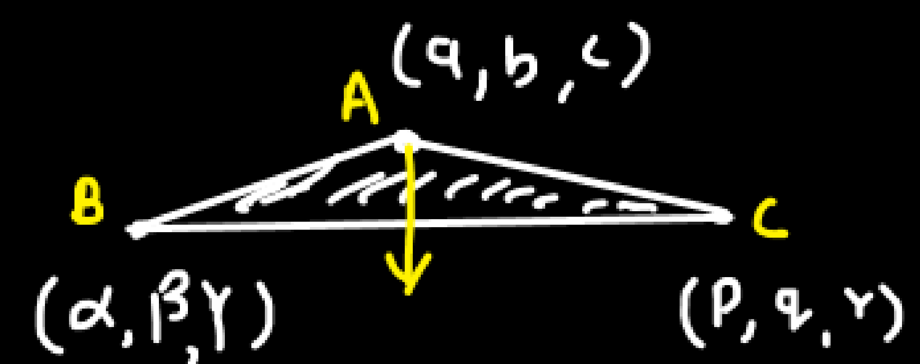
$$= \frac{1}{2}\begin{vmatrix} 3 & 8 & 1 \\ -7 & -6 & 0 \\ 2 & -7 & 0 \end{vmatrix} = \frac{1}{2}[49+12] = \frac{61}{2} \text{ unit}^2$$

Ans =

Q. Find the value of k such that (2, 1), (4, -1) & (k, 0) are co-linear.

↳





$$D = \frac{1}{2}\begin{vmatrix} 2 & 1 & 1 \\ 4 & -1 & 1 \\ k & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow k(2) + 1(-6) = 0$$

$$\boxed{k = 3}$$

* Finding Adjoint of a Matrix :- [Square Matrices]

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}_{3\times 3}$$

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

$$\boxed{|A| = 4}$$

Matrices

$$Adj(A) = \begin{bmatrix} -1 & 2 & -1 \\ 2 & 0 & 2 \\ 1 & -2 & -3 \end{bmatrix}^T = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & -3 \end{bmatrix}_{3\times 3}$$

$$\underbrace{A \cdot \operatorname{adj}(A)}_{\text{Matrix}} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3} = 4 I_3$$

$$= |A| \, I_3$$

AK

$$\boxed{A \cdot \operatorname{adj}(A) = |A| \, I_n}$$

★ Left Multiplication and Right Multiplication of Matrix:-

↳ Given $AB = X$

[Pre] Left Multiplication:-

$AB = X$ ✓

$$\boxed{CAB = CX}$$

$$\boxed{CAB \neq XC}$$

[Post] Right Multiplication:-

$AB = X$

$$\boxed{ABD = XD}$$

$$\boxed{ABD \neq DX}$$

$$\boxed{AB = 5P}$$

$CAB = C[5P] = 5CP$

$$\boxed{CAB = 5CP}$$

$$\boxed{ABD = 5PD}$$

* Finding Inverse of a Matrix A :- [Square Matrices]

Let Matrix $A = \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}$; $B = \begin{bmatrix} 0.4 & -0.2 \\ 0.6 & 0.2 \end{bmatrix}$

calculate $AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; $BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $AB = I = BA$

"If for a given square matrix A, there exists another matrix B such that"

$$AB = I \quad \text{and} \quad BA = I$$

"then the matrix B is called the **inverse of matrix** A, and it is denoted by"

$$B = A^{-1}$$

"Equivalently, we can also say that A is the inverse of B, and we write"

$$A = B^{-1}$$

$AB = I$; $BA = I$

Left Multiplication of A^{-1} ; Left Multiplication of B^{-1}

$$A^{-1}AB = A^{-1}I$$

$$IB = A^{-1}I$$

$$\boxed{B = A^{-1}}$$

$$B^{-1}BA = B^{-1}I$$

$$IA = B^{-1}I$$

$$\boxed{A = B^{-1}}$$

$$\boxed{AA^{-1} = I = A^{-1}A}$$

Finding Inverse of Matrix A:-

$$\boxed{A \cdot adj(A) = |A| I}$$

$$A^{-1}A \cdot adj(A) = |A| A^{-1} I$$

$$I \cdot adj(A) = |A| A^{-1}$$

$$\boxed{adj(A) = |A| A^{-1}}$$

$$\boxed{A^{-1} = \frac{adj(A)}{|A|}}$$

Q. Given $A = \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}_{2\times 2}$; Find A^{-1} ? ; $|A| = 5$

$\begin{matrix} + & - \\ - & + \end{matrix}$

↳ $A^{-1} = \dfrac{adj(A)}{|A|} = \dfrac{1}{|A|} \times adj(A)$

$$\boxed{A^{-1} = \frac{1}{5}\begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}}$$

$$adj(A) = \begin{bmatrix} 2 & 3 \\ -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \times \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 0.4 & -0.2 \\ 0.6 & 0.2 \end{bmatrix} = A^{-1}}$$

Shortcut for 2x2 Matrix:-

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2\times 2} \; ; \; A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Q.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \; ; \; A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$A^T = B$

B^{-1}

$|A| = 4$

$$C = A^{-1} = \frac{1}{4}\begin{bmatrix} -1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & -3 \end{bmatrix}_{3\times 3}$$

C^T

$$\boxed{C^T = B^{-1}}$$

★ **Important Properties :-**

$$\boxed{(A^T)^n = (A^n)^T}$$

$$\boxed{|A^n| = |A|^n}$$

1. $A \cdot A^{-1} = I = A^{-1} A$
2. $(A^T)^{-1} = (A^{-1})^T$
3. $(A^\theta)^{-1} = (A^{-1})^\theta$
4. $(ABCD)^{-1} = D^{-1} C^{-1} B^{-1} A^{-1}$
5. $\text{Det}[A^{-1}] = |A^{-1}| = 1/|A| = [|A|]^{-1} = \dfrac{1}{|A|}$
6. $A \cdot \text{adj}(A) = |A| I$
7. $(A^{-1})^{-1} = A$
8. $(KA)^{-1} = K^{-1} A^{-1} = \dfrac{A^{-1}}{K}$

$$\boxed{\begin{aligned} A \cdot A^{-1} &= I \\ |A \cdot A^{-1}| &= |I| \\ |A| \cdot |A^{-1}| &= 1 \\ |A^{-1}| &= \frac{1}{|A|} \end{aligned}}$$

★ **Some Basic Properties of Adjoint :-**

1. $\text{Adj}(A^T) = [\text{Adj}(A)]^T$
2. $\text{Adj}(A^n) = [\text{Adj}(A)]^n$
3. $\text{Adj}(ABCD) = \text{Adj}(D) \cdot \text{Adj}(C) \cdot \text{Adj}(B) \cdot \text{Adj}(A)$

Q. Matrix A is of order $n \times n$. Its adjoint is $Adj(A)$.

Correct statement?

(a) $Adj[Adj\,A] = |A|^{n-2} \cdot A$ (b) $|Adj\,A| = |A|^{n-1}$

(c) $|Adj[Adj\,A]| = |A|^{(n-1)^2}$ (d) $Adj(kA) = k^{n-1} \cdot Adj(A)$

↳ (a) $A \xrightarrow{Adj} \underset{=B}{Adj(A)} \xrightarrow{Adj} Adj(B) = Adj[Adj(A)]$

$$\boxed{B \cdot Adj(B) = |B| I_n}$$

↳ $Adj(B) = |B| B^{-1}$

$= |Adj\,A| (Adj\,A)^{-1}$

$Adj[Adj(A)] = |A|^{n-1} \cdot Adj[A^{-1}]$ — (1)

$= |A|^{n-1} \cdot \frac{1}{|A|} AI = \boxed{|A|^{n-2} \cdot A}$

$A^{-1} \cdot adj(A^{-1}) = |A^{-1}| \cdot I$

$A A^{-1} \cdot adj(A^{-1}) = |A^{-1}| \cdot AI$

$adj(A^{-1}) = \frac{1}{|A|} A$ — (2)

$$|\mathrm{Adj}(A)| = ?$$

(b) $A\ \mathrm{adj}(A) = |A|\, I_n$

$$A \cdot \mathrm{Adj}(A) = K I_n \quad ; \quad K = |A|$$

$$|A \cdot \mathrm{Adj}(A)| = |K I_n|$$

$$|A|\, |\mathrm{Adj}(A)| = K^n |I_n| \qquad \{ |AB| = |A|\,|B| \}$$

$$|A| \cdot |\mathrm{Adj}(A)| = |A|^n \qquad \{ |I_n| = 1 \}$$

$$|\mathrm{Adj}(A)| = \frac{|A|^n}{|A|}$$

$$\boxed{|\mathrm{Adj}(A)| = (|A|)^{n-1}}$$

(c) $|\text{Adj}\,[\underline{\text{Adj}\,A}]| = |\text{Adj}(A)|^{n-1}$ $\qquad |\text{Adj}(\underline{B})| = |\underline{B}|^{n-1}$

$= (K)^{n-1}$

$K = |\text{Adj}\,A|$ $\qquad = [|A|^{n-1}]^{n-1}$

$K = |A|^{n-1}$ $\qquad = [|A|]^{(n-1)(n-1)}$

$= |A|^{(n-1)^2}$

$|\text{Adj}\,[\text{Adj}\{\text{Adj}(A)\}]| = |A|^{(n-1)^3}$

$|\text{Adj}\,[\text{Adj}\,\underbrace{-\,-\,-\,-}_{m\,\text{times}}\,(A)]| = |A|^{(n-1)^m}$

(d) $\operatorname{adj}(kA) = |kA|\,(kA)^{-1}$

$$= k^n |A| \frac{A^{-1}}{k}$$

$$= k^{n-1} |A| A^{-1}$$

$$\boxed{\operatorname{adj}(kA) = k^{n-1} \operatorname{adj}(A)}$$

$$\operatorname{adj}(-A) = (-1)^{n-1} \operatorname{adj}(A)$$

if $n =$ Even $\Rightarrow$ $\operatorname{adj}(-A) = -\operatorname{adj}(A)$

$n =$ odd $\Rightarrow$ $\operatorname{adj}(-A) = \operatorname{adj}(A)$

$B \cdot \operatorname{adj}(B) = |B| I$

$\operatorname{adj}(B) = |B| B^{-1}$

Q.

Let A be a square matrix of order $n \times n$. Which of the following statements are true?

(A) If A is symmetric, then $\mathrm{adj}(A)$ is always symmetric.
(B) If A is skew-symmetric, then $\mathrm{adj}(A)$ is skew-symmetric if n is even.
(C) If A is skew-symmetric, then $\mathrm{adj}(A)$ is symmetric if n is odd.
(D) All of the above.

(A) Given :- $A^T = A$

$B = \mathrm{adj}(A)$

$B^T = [\mathrm{adj}(A)]^T$

$= \mathrm{adj}(A^T)$

$B^T = \mathrm{adj}(A)$

$\boxed{B^T = B}$ → Symm

(B) (C) : $A^T = -A$

$B = \mathrm{adj}(A)$

$B^T = \mathrm{adj}(A^T)$

$\boxed{B^T = \mathrm{adj}(-A)}$?

→ skew

$B^T = -\mathrm{adj}(A) = -B$; $n = \text{Even}$

$B^T = \mathrm{adj}(A) = B$; $n = \text{odd}$

→ symm

$$\boxed{B \cdot adj(B) = |B|\, I}$$

? $adj(-A) = ?$

$$adj(-A) = |-A|\,(-A)^{-1}$$

$$= |(-1)(A)|\,(-1)\,A^{-1}$$

$$= (-1)^{n}\,|A|\,(-1)\,A^{-1}$$

$$= (-1)^{n+1}\,|A|\,A^{-1}$$

$$\boxed{adj(-A) = (-1)^{n+1}\,adj(A)}$$

$n = \text{Even}$: $adj(-A) = -\,adj(A)$

$n = \text{odd}$: $adj(-A) = adj(A)$

$=$

↳ **Important Point:-**

Singular Matrix:
A square matrix A is called **singular** if

$$\det(A) = 0$$

(It does **not** have an inverse.)

$|A| = 0$

Non-Singular Matrix:
A square matrix A is called **non-singular** if

$$\det(A) \neq 0$$

(It **has** an inverse.)

$|A| \neq 0$

If A is a singular Matrix
⇓
Its Inverse $[A^{-1}]$
Doesn't exist.

$$A^{-1} = \frac{adj(A)}{|A|}$$; if A is singular Matrix → $|A| = 0$

$\boxed{A^{-1} = \infty}$ → A^{-1} Doesn't exist

Q. $[A] = i^2 - j^2$; $[A] = [a_{ij}]$

Comment on [A]. Find Determinant of [A]

(a) If order of [A] is 3x3.

(b) If order of [A] is 2x2.

↳

$$[A] = \begin{bmatrix} 0 & -3 & -8 \\ 3 & 0 & -5 \\ 8 & 5 & 0 \end{bmatrix}_{3\times 3}$$

→ skew symmetric Matrix of odd order

→ $\boxed{|A| = 0}$

$$[A] = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}_{2\times 2}$$

→ skew symmetric Matrix of Even order

→ $\boxed{|A| = 9}$

↳ Important Conclusion:-

1. $[A]_{n\times n} = (i)^{\alpha} - (j)^{\alpha}$ $\alpha \in$ Real Integer

then $[A]_{n\times n}$ will be skew symmetric

Eg:- $\left.\begin{array}{l} [A]_{3\times 3} = i^{7} - j^{7} \\ [A]_{4\times 4} = i^{135} - j^{135} \end{array}\right\} \rightarrow$ skew matrix

2. For a skew Symmetric Matrix $[A]_{n\times n}$; $|A| = 0 \rightarrow n:$ odd

$|A| \neq 0 \rightarrow n:$ Even

3. For a skew symmetric Matrix, Summation of all the elements $= 0$

Q. $[A] = [a_{ij}]_{n\times n} = i^6 - j^6$

$[B] = [A]^m = [b_{ij}]_{n\times n}$

correct statement ?

$\sum_j \sum_i a_{ij}$ = summation of all element of Matrix $[A]$

(a) A^{-1} doesn't exist if n = odd

(b) B^{-1} doesn't exist if $n = 113$

(c) $\underbrace{\sum_i \sum_j a_{ij}}_{} = 0 = \underbrace{\sum_i \sum_j b_{ij}}_{}$, if m = odd

(d) $|B| = 0$ if $m = 27$ and $n = 45$

$[A]$: skew symm. Matrix of order n $\quad A^T = -A$

$[B]$ → symm. if m = Even
$[B]$ → skew symm. if m = odd

(a) $[A]$: skew symm. Matrix of odd order $\Rightarrow |A| = 0 \Rightarrow A^{-1}$ D.N.E.

(b) $n = 113$: odd $\quad$ [B] is having odd order

$\downarrow$

$|B| = 0 \rightarrow$ NO

$\downarrow$

[only when B is skew symm]

$|B| = 0$ or B^{-1} D.N.E. if $\underbrace{m = \text{odd}}_{B:\ \text{skew symm.}}$ and $\underbrace{n = \text{odd}}$

↳ $|B| = 0$

(c) $\displaystyle\sum_{i=1}^{5} a_i = a_1 + a_2 + a_3 + a_4 + a_5$

$\displaystyle\sum_{\substack{i=1 \\ j}}^{5} a_{ij} = a_{1j} + a_{2j} + a_{3j} + a_{4j} + a_{5j}$

↳ If m: odd $\Rightarrow$ [B]: skew symm.

$$\sum_i \sum_j a_{ij} = a_{11} + a_{12} + a_{13} + \cdots$$
$$a_{21} + a_{22} + a_{23} + \cdots$$
$$a_{31} + a_{32} + a_{33} + \cdots$$
$$\vdots \qquad \vdots$$
$$+ a_{\infty 1} + a_{\infty 2} + a_{\infty 3} + \cdots$$

Some Questions :-

Q. $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$, $AA^T = 9I_3$; Find the value of a and b.

$2(1) + 1(2) + (-2)(2) = 0$

$1^2 + 2^2 + (-2)^2 = 9$

$AA^T = I \Rightarrow A$: orthogonal $\quad a^2 + b^2 + c^2 = 1 =$

$AA^T = 9I \Rightarrow a^2+b^2+c^2 = 9, \quad d^2+e^2+f^2 = 9, \quad g^2+h^2+i = 9$

$ad + be + cf = 0, \quad dg + eh + fi = 0 \quad - - -$

$2+2+2a = 0 \Rightarrow \boxed{a = -2}$ $\qquad 2(2) + 1(-2) + 2b = 0 \quad \boxed{b = -1}$

Q.

Question:

Let A and B be two singular matrices such that

$$AB = B \quad \text{and} \quad BA = A.$$

Then, $A^2 + B^2$ is equal to:

(A) $A + B$

(B) AB

(C) BA

(D) $2(A + B)$

$A^2 + B^2 = A \cdot A + B \cdot B$

$= \underbrace{A \cdot B}A + \underbrace{B \cdot A}B$

$= BA + AB$

$= A + B$

Q.

Question:

Find the number of different **symmetric** 4×4 matrices in which each element is either 2 or 3.

↳

$$A = \begin{bmatrix} a & e & f & h \\ \cancel{e} & b & g & i \\ \cancel{f} & & c & j \\ k & & & d \end{bmatrix}$$

$$(2)^{10} = \boxed{1024}$$

↳ Number of different $n \times n$ symm. matrices in which each element is either 0 or 1 $= 2^{n(n+1)/2}$

↳ " " " $n \times n$ skew sym " " $= 2^{n(n-1)/2}$

Q.

Question:

Let S be the set of all 6×6 matrices $A = [a_{ij}]$ such that

- $a_{ij} \in \{0, 1\}$ for all i, j, → all elements are either 1 or 0.
- For each column j: $\sum_{i=1}^{6} a_{ij} = 1$, → sum of all the element in each column is 1.
- For each row i: $\sum_{j=1}^{6} a_{ij} = 1$. ↳ " " " " " " " row is 1.

Find the number of elements in the set S.

↳

$S : A = [a_{ij}]$

$$\sum_{i=1}^{6} a_{ij} = 1 = a_{1j} + a_{2j} + a_{3j} + a_{4j} + a_{5j} + a_{6j}$$

if $j = 1$ →

$$a_{11} + a_{21} + a_{31} + a_{41} + a_{51} + a_{61} = 1$$

$$
\begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}_{6\times 6}
\qquad
\begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}_{6\times 6}
$$

$$
\begin{array}{l}
\Sigma = 1 \\
\Sigma = 1 \\
\\
\\
\\
\\
\end{array}
\begin{bmatrix}
\underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{0} \\
\underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{0} \\
\_ & \underline{0} & 0 & \_ & \_ & \_ \\
 & & 0 & & & \\
 & & 0 & & & \\
 & & 0 & & &
\end{bmatrix}
\begin{array}{l}
\to 6 \\
\to 5 \\
\to 4 \\
\to 3 \\
\to 2 \\
\to 1
\end{array}
$$

$$
6 \times 5 \times 4 \times 3 \times 2 \times 1 = \boxed{6!} = \boxed{720}
$$

Q.

Question:

Let A be a 3×3 matrix such that $\det(A) = -2$. Define $B = \mathrm{adj}(A)$.

Find the value of

$$\det(AB + 4I),$$

where I is the identity matrix of order 3.

↳

$$|AB + 4I| = |A \cdot \mathrm{adj}\, A + 4I|$$

$$= |-2I_3 + 4I_3| = |2I_3|$$

$$= (2)^3 |I_3|$$

$$= \boxed{8}$$

ANS.

$$A \cdot \mathrm{adj}(A) = |A| I$$

$$= -2I$$

Q. $A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$

Given $A^2 - 6A + 9I_2 = [0]$; Find $A^{-1} = ?$

$A - 6I + 9A^{-1} = 0$

$A^{-1} = \dfrac{6I - A}{9}$

$$\boxed{A^{-1} = \frac{1}{9}\begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}}$$

Q. A is 3×3 Matrix with $|A| = -2$, then

(a) $Adj[Adj(A)] = \alpha A$; Find α ; Ans. $= -2$

(b) $|Adj[Adj[Adj(A)]]| = \beta$; Find β ; Ans. $= 256$

(c) $Adj(-3A) = \gamma\, Adj(A)$; Find γ ; Ans. $= 9$

(d) $Adj[Adj(-3A)] = \eta\, Adj[Adj(A)]$; Find η ; Ans. $= 81$

↳ $\boxed{A \cdot adj(A) = |A| I}$ $\qquad |adj\, A| = |A|^{n-1}$

$$A \cdot adj\, A = |A| I$$

$$|A| \cdot |adj\, A| = \big|\, |A| I \,\big|$$

$$|A| \cdot |adj\, A| = |A|^3$$

$$|adj\, A| = |A|^2$$

(a) $adj(A) = |A| A^{-1}$

$$adj[adj\, A] = |adj\, A| \cdot [adj(A)]^{-1}$$

$$= |A|^2 \cdot [adj(A^{-1})] \quad — (1)$$

$$\text{adj}(A^{-1}) = |A^{-1}|\,(A^{-1})^{-1}$$

$$\boxed{\text{adj}(A^{-1}) = \frac{1}{|A|}A}$$

$$\text{adj}(\text{adj}\,A) = |A|^2\,\frac{A}{|A|} = \underbrace{|A|}_{\alpha}\cdot A$$

$$\boxed{\alpha = |A| = -2}$$

(b) $$|\text{adj}[\text{adj}\{\text{adj}(A)\}]| = |A|^{(n-1)^3}$$

$$= (-2)^{2^3} = (-2)^8 = 256 = \text{ANS.}$$

(c) $$\text{adj}(kA) = k^{n-1}\,\text{adj}(A)$$

$$\text{adj}(-3A) = (-3)^2\,\text{adj}(A) = \boxed{9}\,\text{adj}(A)$$

(a) $\left[Adj\left[Adj(-3A)\right]\right] = ?$

$$= (-3)^{2^2} Adj\left[Adj\, A\right]$$

$$= (-3)^4 Adj\left[Adj\, A\right]$$

$$= \boxed{81}\ adj\left[adj\, A\right]$$

$$Adj(kA) = k^{n-1} Adj(A)$$

$$Adj\left[Adj(kA)\right] = Adj\left[k^{n-1}\, Adj\, A\right]$$

$$= Adj\left[\alpha B\right]$$

$$= \alpha^{n-1}\, Adj(B)$$

$$= (k)^{(n-1)^2} Adj\left[Adj\, A\right]$$

Sub-Matrices:-

A submatrix of a matrix is obtained by deleting one or more rows and/or one or more columns from the original matrix.

Eg. →

$$A = \begin{array}{c} \begin{array}{cccc} c_1 & c_2 & c_3 & c_4 \end{array} \\ \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 0 & -1 & -2 \end{bmatrix} \end{array} \begin{array}{c} R_1 \\ R_2 \\ R_3 \end{array}_{3\times 4}$$

; Write all possible square sub-Matrices.

↳ 4x4, 5x5, 6x6 submatrices --- are not possible.

3x3 sub-matrices :- $^3c_3 \times {^4c_3} = 4$

$$\begin{array}{c|c} & \begin{array}{ccc} c_1 & c_2 & c_3 \end{array} \\ \begin{array}{c} R_1 \\ R_2 \\ R_3 \end{array} & \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 9 & 0 & -1 \end{bmatrix} \end{array} ; \quad \begin{array}{c} \begin{array}{ccc} c_2 & c_3 & c_4 \end{array} \\ \begin{bmatrix} 2 & 3 & 4 \\ 6 & 7 & 8 \\ 0 & -1 & -2 \end{bmatrix} \end{array} ; \quad \begin{array}{c} \begin{array}{ccc} c_1 & c_2 & c_4 \end{array} \\ \begin{bmatrix} 1 & 2 & 4 \\ 5 & 6 & 8 \\ 9 & 0 & -2 \end{bmatrix} \end{array} ; \quad \begin{array}{c} \begin{array}{ccc} c_1 & c_3 & c_4 \end{array} \\ \begin{bmatrix} 1 & 3 & 4 \\ 5 & 7 & 8 \\ 9 & -1 & -2 \end{bmatrix} \end{array}$$

2x2 sub-matrices:-

$$\underset{\hookrightarrow p}{{}^{3\,\nearrow m}C_2} \times \underset{\hookrightarrow p}{{}^{4\,\nearrow n}C_2} = 3 \times 6 = 18$$

$$\begin{bmatrix}1 & 2\\5 & 6\end{bmatrix}, \begin{bmatrix}1 & 3\\5 & 7\end{bmatrix}, \begin{bmatrix}1 & 4\\5 & 8\end{bmatrix}$$

$$\begin{bmatrix}2 & 3\\6 & 7\end{bmatrix}, \begin{bmatrix}2 & 4\\6 & 8\end{bmatrix}, \begin{bmatrix}3 & 4\\7 & 8\end{bmatrix}$$

$$\begin{bmatrix}1 & 2\\9 & 0\end{bmatrix}, \begin{bmatrix}1 & 3\\9 & -1\end{bmatrix}, \begin{bmatrix}1 & 4\\9 & -2\end{bmatrix}$$

$$\begin{bmatrix}2 & 3\\0 & -1\end{bmatrix}, \begin{bmatrix}2 & 4\\0 & -2\end{bmatrix}, \begin{bmatrix}3 & 4\\-1 & -2\end{bmatrix}$$

$$\begin{bmatrix}5 & 6\\9 & 0\end{bmatrix}, \begin{bmatrix}5 & 7\\9 & -1\end{bmatrix}, \begin{bmatrix}5 & 8\\9 & -2\end{bmatrix}$$

$$\begin{bmatrix}6 & 7\\0 & -1\end{bmatrix}, \begin{bmatrix}6 & 8\\0 & -2\end{bmatrix}, \begin{bmatrix}7 & 8\\-1 & -2\end{bmatrix}$$

1x1 Sub-Matrices:-

$${}^{3}C_1 \times {}^{4}C_1 = 3 \times 4 = 12$$

$$[1],\ [2],\ [3],\ [4],\ [5],\ [6],\ [7],\ [8],\ [9],\ [0],\ [-1],\ [-2]$$

↳ For a $\overset{R}{m} \times \overset{C}{n}$ matrix ; maximum possible square sub-matrix of order p is = ${}^{m}C_p \times {}^{n}C_p$

Q. From a 4x5 matrix ; How many 3x3 square sub-matrices you can make?

↳ 5x5 sub matrices = X

4x4 sub matrices = ${}^{4}C_4 \times {}^{5}C_4 = 5$

3x3 " " $= {}^{4}C_3 \times {}^{5}C_3 = 4 \times 10 = 40$

2x2 " " $= {}^{4}C_2 \times {}^{5}C_2 = 6 \times 10 = 60$

1x1 " " $= {}^{4}C_1 \times {}^{5}C_1 = 4 \times 5 = 20$

"At least one Minor $\neq 0$"

* Rank of a Matrix :-

"Rank is the size of the largest square submatrix whose determinant is non-zero."

Minor

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 1 & 0.5 & -0.5 \\ 1 & 0 & -1 & 2 \end{bmatrix}_{3\times4}$$

3x3 : Highest order

2x2 :

1x1 :

$\boxed{\text{Rank} = 3}$

3x3 sub-matrices :-

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0.5 \\ 1 & 0 & -1 \end{bmatrix} ; \quad \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0.5 & -0.5 \\ 0 & -1 & 2 \end{bmatrix} ; \quad \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -0.5 \\ 1 & -1 & 2 \end{bmatrix} ; \quad \begin{bmatrix} 1 & 1 & -1 \\ 0 & 0.5 & -1/2 \\ 1 & -1 & 2 \end{bmatrix}$$

$\Downarrow$ Det $\neq 0$

$R_2 \to R_2 - R_1/2$ $\Downarrow$ Det $= 0$

$\Downarrow$ Det $\neq 0$

$\Downarrow$ Det $\neq 0$

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0.5 & 1 & 0.5 & -0.5 \\ 1 & 0 & -1 & 2 \end{bmatrix}_{3\times 4}$$

$\hookrightarrow 3\times3$ ✗

$\hookrightarrow 2\times2$ ✓

$\hookrightarrow 1\times1$

$\boxed{\text{Rank} = 2}$

↳ <u>All 3x3 Matrices:-</u>

$$\begin{bmatrix} 1 & 2 & 1 \\ 0.5 & 1 & 0.5 \\ 1 & 0 & -1 \end{bmatrix}_{3\times3}, \quad \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0.5 & -0.5 \\ 0 & -1 & 2 \end{bmatrix}_{3\times3}, \quad \begin{bmatrix} 1 & 2 & -1 \\ 0.5 & 1 & -0.5 \\ 1 & 0 & 2 \end{bmatrix}_{3\times3}, \quad \begin{bmatrix} 1 & 1 & -1 \\ 0.5 & 0.5 & -0.5 \\ 1 & -1 & 2 \end{bmatrix}_{3\times3}$$

Det = 0   Det = 0   Det = 0   Det = 0

<u>All 2x2 sub·matrices:-</u> $^3C_2 \times {^4C_2} = 3\times6 = 18$

$$\begin{bmatrix} 1 & 2 \\ 0.5 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 \\ 0.5 & -0.5 \end{bmatrix} \quad \begin{bmatrix} 0.5 & 1 \\ 1 & 0 \end{bmatrix}$$

Det = 0   Det = 0   Det = $-1 \neq 0$ **

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0.5 & 1 & 0.5 & -0.5 \\ 2 & 4 & 2 & -2 \end{bmatrix}_{3\times 4}$$

$\boxed{\text{Rank} = 1}$

<u>all 3x3 subMatrices:-</u> Det = 0

<u>all 2x2 sub matrices:-</u> Det = 0

<u>1x1 sub matrices:-</u>

$[1]$

$\Downarrow$

$\text{Det} = 1 \neq 0$

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4}$$

Rank = 0 = undefined

All 3x3 submatrices :- Det = 0

All 2x2 submatrices :- ''

All 1x1 submatrices :- ''

positive Integer

- Rank (A) ≥ 0
- If Rank(A) = 0

$\Downarrow$

[A] = Null Matrix

$$\boxed{\text{Maximum Rank of } [A]_{m\times n} = \min[m, n]}$$

Possible

Q. Find maximum possible rank.

Rank [A] = ρ[A] ↳ Rho

(a)
$$\begin{bmatrix} a & b & c & d & m \\ e & f & g & h & \alpha \\ i & j & k & l & \beta \end{bmatrix}_{3\times 5} \Rightarrow \rho_{max} = 3$$

(b)
$$\begin{bmatrix} a & e \\ b & f \\ c & g \\ d & h \end{bmatrix}_{4\times 2} \Rightarrow \rho_{max} = 2$$

(c)
$$\begin{bmatrix} a & b & c & d & e \end{bmatrix}_{1\times 5} \quad \rho_{max} = 1$$

(d)
$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}_{4\times 1} \Rightarrow \rho_{max} = 1$$

N.B.:- Minor of a sub-matrix = Determinant of sub-Matrix

Q. **Question:**

Let A be an $m \times n$ matrix with $m < n$, and $A \neq 0$ → Non-Null Matrix. All minors of order m, $(m-1)$, $(m-2)$, and $(m-3)$ are zero, and at least one minor of order $(m-4)$ is non-zero. If the rank of A is r, which of the following statements are true?

(A) $r = m - 4$

(B) $m \neq 4$

(C) $m \geq 5,\ m \in \mathbb{Z}$

(D) All of the above

↳ $[A]_{m \times n}$ $\quad m < n$

Non Null Matrix

$r = m - 4$; $r > 0$

$m - 4 > 0 \rightarrow m > 4 \Rightarrow m \geq 5$

$m \times n \rightarrow$
- $m \times m$: order m
- $(m-1) \times (m-1)$: order $(m-1)$
- $(m-2) \times (m-2)$:

Q. Square Matrix of order $n \times n$ is reversible. Rank = ?

~~(a) n~~ (b) $n-1$ (c) 1 (d) 0

↳ $[A]_{n \times n}$: Reversible $\Rightarrow |A| \neq 0$; Rank $= n$

$A^{-1} = \dfrac{adj(A)}{|A|}$

exist

↳ $|A| \neq 0$

→ $n \times n$: Det $\neq 0$
→ $(n-1) \times (n-1)$
→
⋮

Q. Square Matrix of order $n \times n$ is not reversible. Possible rank = ?

~~(a) n~~ ~~(b) $n-1$~~ ~~(c) 1~~ ~~(d) $n-7$~~

↳ $[A]_{n \times n}$: NOT Reversible $\Rightarrow |A| = 0$; Rank = ?

↳ Minor of $n \times n = 0$

→ $n \rightsquigarrow$ Det $= 0$
→ $n-1 \rightsquigarrow$ Det $\neq 0$
→ $n-2 \rightsquigarrow$ --- $n-7$ --- 2, 1.

$$\boxed{\begin{array}{l} \text{Rank} < n \\ \text{Rank} \leq n-1 \end{array}}$$

Understanding Echelon (Row Echelon) form of a Matrix:-

Echelon Form (Easy Language)

1. The number of zeros before the first non-zero element in a row is less than the number of zeros before the first non-zero element in the next row.
2. All zero rows, if any, must be at the bottom (after all non-zero rows).

One-line version: Each row starts further right, and zero rows are at the bottom.

Eg. -

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 6 & 7 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{array}{c} \#\text{zero} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \quad ; \quad B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \#\text{zero} \\ 0 \\ 2 \\ 3 \\ \text{All zero} \end{array}$$

R.E. ✓ R.E. ✓

$$\boxed{\rho = 4} \qquad \boxed{\rho = 3}$$

$$C = \begin{array}{c} 4 \\ 4 \\ 4 \\ 4 \end{array}\begin{bmatrix} 0 & 2 & 4 & 3 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \#\text{zero} \\ 1 \\ 2 \\ 3 \\ \text{All zero} \end{array} \quad ; \quad D = \begin{array}{c} 4 \\ 4 \\ 4 \\ 4 \end{array}\begin{bmatrix} 0 & 2 & 4 & 3 \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \#\text{zeros} \\ 1 \\ 2 \\ 2 \\ \text{All zero} \end{array}$$

Equal

R.E. ✓ $\qquad$ R.E. ✗

$$\boxed{\rho = 3}$$

$$E = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 6 & 7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 4 \end{bmatrix} \begin{array}{l} \#\text{zero} \\ \to 0 \\ \to 1 \\ \to \text{All zero} \\ \to 2 \end{array} \quad ; \quad F = \begin{bmatrix} 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \#\text{zero} \\ \to 2 \\ \to 3 \\ \to \text{All zero} \\ \to \text{All zero} \end{array}$$

R.E. ✗ $\qquad$ R.E. ✓

$$\boxed{\rho = 2}$$

$\boxed{\rho = 3}$

$$G = \left[\begin{array}{cccc} 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \to 1 \\ \to 2 \\ \to 3 \\ \to \text{All zero} \end{array}$$

R.E. ✓

$\boxed{\rho = 3}$

$$H = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \# \text{zero} \\ \to 0 \\ \to 1 \\ \to 2 \\ \to \text{All zero} \end{array}$$

R.E. ✓

↳ Getting Echelon (Row Echelon) form of a Matrix:-

1. All Linear Row*** Transformation allowed.

$R_3 \to R_3 - 2R_1 + 4R_2$ ✓

$R_2 \to 4R_2 - 3R_1 + 7R_3$ ✓

$R_1 \to -3R_1$ ✓

$C_1 \to C_1 - 2C_2$ X X

x *column xx

2. Allowed Simulatenous operation :-

(a) $R_2 \to 3R_2 - 4R_1$ ✓
$R_3 \to R_3 - 2R_1$

(b) $R_2 \to 4R_2 - 3R_1$ ✓
$R_1 \to 3R_1 - 2R_3$

3. ✓ Interchanging Rows is allowed. [But not columns]

4. $\left.\begin{aligned} R_3 &\to R_3 - R_2 \\ R_2 &\to R_2 + 2R_3 \end{aligned}\right\}$ ✓ $\Rightarrow$ allowed in R.E.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_2]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$$

$\Downarrow$

NOT Allowed for calculating determinant

↳ No. of Non-zero Rows in Echelon form = Rank

Q. Find Rank.

(a)

$$\begin{array}{c} 0 \leftarrow \\ 1 \leftarrow \\ 2 \leftarrow \\ 2 \leftarrow \end{array} \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 1 \end{bmatrix} \xrightarrow{R_4 \to 2R_4 - 3R_3} \begin{array}{c} \\ \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} \end{array} \begin{array}{c} \#\text{zero} \\ \to 0 \\ \to 1 \\ \to 2 \\ \to 3 \end{array}$$

R.E. ✗   R.E. ✓

$\#$ non-zero Rows = $\boxed{4 = \text{Rank}}$

(b)

$$\begin{array}{r} 2^{nd} \to 1 \\ 3^{rd} \to \text{All } 0 \to \\ 1^{st} \to 0 \leftarrow \end{array} \begin{bmatrix} 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 3 & 4 \end{bmatrix} \xrightarrow[\text{Rows}]{\text{Interchanged}} \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \#\text{Zeros} \\ \to 0 \\ \to 1 \\ \to \text{All zero} \end{array}$$

R.E. ✗   R.E. ✓

$\boxed{\text{Rank} = 2}$

#

(c)
$$\begin{array}{c} 0\\0\\0\\0 \end{array}\begin{bmatrix} 1 & 3 & -1 & 4 \\ 2 & 4 & 2 & 6 \\ 1 & 5 & 0 & 2 \\ 3 & 6 & 3 & 3 \end{bmatrix} ✓ \xrightarrow[\substack{R_3 \to R_3 - R_1 \\ R_4 \to R_4 - 3R_1}]{R_2 \to R_2 - 2R_1} \begin{array}{c} \begin{bmatrix} 1 & 3 & -1 & 4 \\ 0 & -2 & 4 & -2 \\ 0 & 2 & 1 & -2 \\ 0 & -3 & 6 & -9 \end{bmatrix} \end{array}\begin{array}{c} \to 0 \\ \to 1 \\ \to 1 \\ \to 1 \end{array}$$

R.E. ✗

$$\Big\downarrow \; \substack{R_3 \to R_3 + R_2 \\ R_4 \to 2R_4 - 3R_2}$$

$$\begin{bmatrix} 1 & 3 & -1 & 4 \\ 0 & -2 & 4 & -2 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 0 & -12 \end{bmatrix}\begin{array}{c} \to 0 \\ \to 1 \\ \to 2 \\ \to 3 \end{array}$$

R.E. ✓

$$\boxed{\text{Rank} = 4}$$

(d)

$$\begin{array}{c} 1 \hookrightarrow \\ 0 \hookrightarrow \\ 0 \hookrightarrow \\ 1 \hookrightarrow \end{array} \begin{bmatrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 3 & 2 \\ 3 & 0 & 9 & 6 \\ 0 & \frac{3}{4} & \frac{3}{2} & -\frac{3}{4} \end{bmatrix} \xrightarrow{\text{Interc.}} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 0 & 9 & 6 \\ 0 & 1 & 2 & -1 \\ 0 & \frac{3}{4} & \frac{3}{2} & -\frac{3}{4} \end{bmatrix} \begin{array}{l} \hookrightarrow 0 \\ \hookrightarrow 0 \\ \hookrightarrow 1 \\ \hookrightarrow 1 \end{array}$$

$R_2 \to R_2 - 3R_1$

$R_4 \to R_4 - \frac{3}{4} R_3$

$$\begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \to 0 \\ \to \text{All zero} \\ \to 1 \\ \to \text{All zero} \end{array}$$

$$\xrightarrow{\text{Interc.}}$$

$$\begin{array}{r} \\ \text{R.E.} \downarrow \\ \hookrightarrow \\ \text{All zero} \hookrightarrow \end{array} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \to 0 \\ \to 1 \\ \\ \\ \end{array}$$

$$\boxed{\text{Rank} = 2}$$

(e)

$$\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\begin{bmatrix} 1 & 2 & 4 & 7 \\ 3 & 2 & 1 & 5 \\ 4 & 4 & 5 & 12 \\ 7 & 6 & 6 & 17 \end{bmatrix} \xrightarrow[\substack{R_3 \to R_3 - (R_1 + R_2) \\ R_4 \to R_4 - (R_2 + R_3)}]{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 2 & 4 & 7 \\ 0 & -4 & -11 & -16 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{array}{l} \to 0 \\ \to 1 \\ \to \text{All zero} \\ \end{array}$$

R.E. ✓

$$\boxed{\text{Rank} = 2}$$

(f)

$$\begin{array}{c} 2 \\ 2 \\ 2 \\ 2 \end{array}\begin{bmatrix} 0 & 0 & 2 & -1 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 6 & -3 \\ 0 & 0 & 3/2 & 2 \end{bmatrix}_{4\times 4}$$

$R_3 \to R_3 - 3R_1$

$R_4 \to R_4 - R_2/2$

$R_2 \to 2R_2 - 3R_1$

$$\begin{bmatrix} 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 11 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{array}{l} \hookrightarrow 2 \\ \hookrightarrow 3 \\ \left.\begin{array}{l} \\ \end{array}\right\} \to \text{All Zero} \end{array}$$

$$\boxed{\text{Rank} = 2}$$

4×4 3×3 2×2 1×1

4×4:

$$\begin{bmatrix} 0 & 0 & & \\ 0 & 0 & & \\ 0 & 0 & & \\ 0 & 0 & & \end{bmatrix}$$

$\Downarrow$

Det $= 0$

Rank $\neq 4$

3×3:

${}^4C_3 \times {}^4C_3 = 16$

$\Downarrow$

Det $= 0$

Rank $\neq 3$

2×2:

$\Downarrow$

Det $\neq 0$

$$\boxed{\text{Rank} = 2}$$

(g)

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix} \xrightarrow[R_1 \to R_1 - 2R_3]{R_2 \to R_2 - (R_1 + R_3)} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

$\big\downarrow$ Interchanged

$$\begin{array}{c} \\ \\ \\ \end{array}\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & 3 \end{bmatrix} \begin{array}{c} \#\text{zeros} \\ 0 \\ 2 \\ 2 \end{array}$$

$$\xrightarrow{R_3 \to R_3 - 3R_2} \begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

R.E. ✓

$\boxed{\text{Rank} = 2}$

$\boxed{\eta(A) = 4 - 2 = 2}$

(4)

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix} \xrightarrow[R_4 \to R_4 - R_3]{R_2 \to R_2 - R_1,\; R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 4 & 9 & 16 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 11 \\ 7 & 9 & 11 & 13 \end{bmatrix}$$

$R_2 \to R_2 - 3R_1$
$R_3 \to R_3 - R_2$
$R_4 \to R_4 - R_3$

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -20 & -39 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

Equivalent Matrix

$R_3 \to R_3 - R_4$

$\underbrace{R_4 \to R_4 - R_3}$

NOT Acceptable

$R_4 \to R_4 - R_3$
$R_3 \to R_3 - 2R_1$

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -20 & -39 \\ 0 & -6 & -16 & -30 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 4}$$

3×3

$[Det] = (16\times 7) - (20\times 6) \neq 0$

$\boxed{\text{Rank} = 3}$

(i)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 2 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_2]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 1 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - 2R_2$

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -4 \\ 0 & -3 & -1 & -1 \end{bmatrix}$$

Rank = 3

$$\begin{array}{c} \#\text{zero} \\ 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & -3 & -1 & -1 \\ 0 & 0 & -3 & -4 \end{bmatrix} \xleftarrow{\text{Inter c.}} \begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -4 \\ 0 & -3 & -1 & -1 \end{bmatrix}$$

$$\boxed{\text{Rank} = 3}$$

$$\boxed{\eta(A) = 4 - 3 = 1}$$

(j)

$$A = \begin{bmatrix} 1 & 3 & 6 & 10 \\ 3 & 6 & 10 & 15 \\ 6 & 10 & 15 & 21 \\ 10 & 15 & 21 & 28 \end{bmatrix} \xrightarrow[\substack{R_3 \to R_3 - R_2 \\ R_4 \to R_4 - R_3}]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 3 & 6 & 10 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix}$$

$$\Big\downarrow \; \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_2 \\ R_4 \to R_4 - R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & 6 & 10 \\ 0 & -3 & -8 & -15 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xleftarrow[\substack{R_3 \to R_3 - R_2 \\ R_4 \to R_4 - R_3}]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 3 & 6 & 10 \\ 1 & 0 & -2 & -5 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$[\text{Det}]_{3\times3} = -1 \neq 0$

$$\boxed{\text{Rank} = 3}$$

$\eta(A) = 4 - 3 = 1$

$\text{Rank} \neq 4$

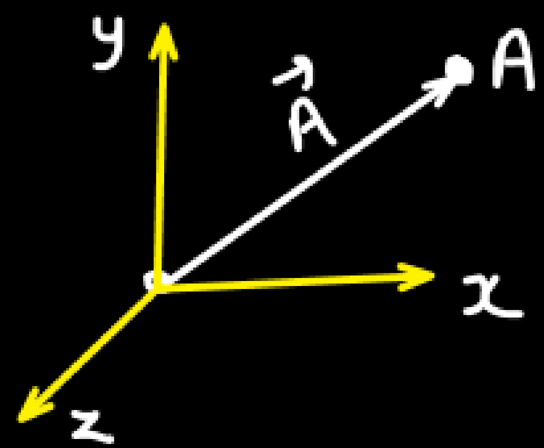
$\text{Rank} = 3\,?$

↳ Relating vector with Matrices:-

① $\vec{A} = 3\hat{i} + 4\hat{j} - 5\hat{k}$

$\rightarrow$ Row vector $= \begin{bmatrix} 3 & 4 & -5 \end{bmatrix}_{1\times 3}$

$\rightarrow$ column vector $= \begin{bmatrix} 3 \\ 4 \\ -5 \end{bmatrix}_{3\times 1}$



② $\vec{B} = 2\hat{i} - 3\hat{j}$

$\rightarrow$ Row vector $= \begin{bmatrix} 2 & -3 \end{bmatrix}_{1\times 2}$

$\rightarrow$ Column vector $= \begin{bmatrix} 2 \\ -3 \end{bmatrix}_{2\times 1}$

* Scalar Multiplication of two vectors:- [DOT Product]

$$\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{P} = p\hat{i} + q\hat{j} + r\hat{k}$$

$$\vec{A}\cdot\vec{P} = ap + bq + cr = \text{Scalar/Constant}$$

Row Matrix representation:-

$$\vec{A} = A = \begin{bmatrix} a & b & c \end{bmatrix}_{1\times 3}$$

$$\vec{P} = P = \begin{bmatrix} p & q & r \end{bmatrix}_{1\times 3}$$

$$\vec{A}\cdot\vec{P} = A\cdot P^T = P\cdot A^T$$

$$A\cdot P^T = \begin{bmatrix} a & b & c \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \underline{ap + bq + cr}$$

Column matrix representation:-

$$\vec{A} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}_{3\times 1} = A \; ; \; \vec{P} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_{3\times 1} = P$$

$$\vec{A}\cdot\vec{P} = ap + bq + cr = \boxed{A^T\cdot P = P^T\cdot A}$$

$$A^T = \begin{bmatrix} a & b & c \end{bmatrix}_{1\times 3} \; ; \; P = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_{3\times 1}$$

$$= \underline{ap + bq + cr}$$

Q. Find the no of linearly dependent and independent vectors?

(a) $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ $\Rightarrow$ $\vec{A} = 2\vec{B}$ OR $\vec{B} = \frac{\vec{A}}{2}$

$\vec{B} = \hat{i} + 1.5\hat{j} - 2\hat{k}$

$\underline{1}$ L.I.D. and $\underbrace{\underline{1} \text{ L.D. vector}}$ set of L.D. vector.

(b) $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ $\boxed{\vec{A} \neq \alpha \vec{B}}$

$\vec{B} = 4\hat{i} + 6\hat{j} - 6\hat{k}$

$\underbrace{\underline{2} \text{ L.I.D. and } \underline{0} \text{ L.D.}}$

set of L.I.D. vector

(c) $\vec{A} = 2\hat{i} + \hat{j} - \hat{k}$

$\vec{B} = 3\hat{i} + \hat{j} - 3\hat{k}$

$\vec{C} = 5\hat{i} + 2\hat{j} - 4\hat{k}$

$\boxed{\vec{A} + \vec{B} = \vec{C}}$ ✓

$\boxed{\vec{C} - \vec{B} = \vec{A}}$

$\boxed{\vec{C} - \vec{A} = \vec{B}}$

$\underline{2}$ L.I.D. and $\underbrace{\underline{1} \text{ L.D.}}_{\text{set of L.D. vector.}}$

(d) $\vec{A} = 2\hat{i} + \hat{j} - \hat{k}$

$\vec{B} = 3\hat{i} + \hat{j} - 3\hat{k}$

$\vec{C} = 4\hat{i} + 2\hat{j} - 4\hat{k}$

$\boxed{\alpha\vec{A} + \beta\vec{B} \neq \vec{C}}$

$\underbrace{\underline{3} \text{ L.I.D. and } \underline{0} \text{ L.D.}}_{\text{set of L.I.D. vector.}}$

(e) $\vec{A} = 2\hat{i} + 4\hat{j} - 3\hat{k}$

$\vec{B} = \hat{i} + 2\hat{j} - 1.5\hat{k}$

$\vec{C} = 4\hat{i} + 8\hat{j} - 6\hat{k}$

$\vec{A} = 2\vec{B}$

$\vec{B} = \frac{1}{4}\vec{C}$

$\vec{C} = 2\vec{A}$

$$\boxed{\begin{aligned}\vec{B} &= \alpha\,\vec{C}\\ \vec{B} &= \beta\,\vec{A}\end{aligned}}$$

$\underline{1}$ L.I.D. $\underbrace{\text{and } \underline{2} \text{ L.D.}}_{\text{set of L.D. vector}}$

↳ <u>Meaning of:-</u>

* "set of Linearly Independent Vectors" → All the vectors should be L.I.D.

* "set of Linearly dependent Vectors" → At least one vector should be L.D.

- vector $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$ and $\vec{E}$ are set of linearly Independent vectors; then

$$\boxed{\lambda_1 \vec{A} + \lambda_2 \vec{B} + \lambda_3 \vec{C} + \lambda_4 \vec{D} + \lambda_5 \vec{E} \neq \vec{0}}$$

where $|\lambda_1| + |\lambda_2| + |\lambda_3| + |\lambda_4| + |\lambda_5| \neq 0$

$\Downarrow$

$\lambda_1, \lambda_2, \lambda_3, \lambda_4$ and λ_5 are simultaneously not zero.

If they were dependent:-

$$\lambda_1 \vec{A} + \lambda_2 \vec{B} + \lambda_3 \vec{C} + \lambda_4 \vec{D} + \lambda_5 \vec{E} = \vec{0}$$

* what rank represents?

↳ Rank tell the no of independent rows (or column).

(a) $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ $\quad \vec{A} = 2\vec{B}$

$\vec{B} = \hat{i} + 1.5\hat{j} - 2\hat{k}$

$\underline{1}$ L.I.D. and $\underline{1}$ L.D.

"Row-wise" Matrix representation:-

$$\begin{array}{c} \vec{A} = \\ \vec{B} = \end{array} \begin{bmatrix} 2 & 3 & -4 \\ 1 & 1.5 & -2 \end{bmatrix} \xrightarrow{\text{Rank}} \underbrace{\rho = 1}$$

1 I.D. Row.

1 I.D. Col.

"Column-wise" Matrix representation:-

$$\begin{array}{cc} & \begin{array}{cc} C_1 & C_2 \end{array} \\ \begin{array}{c} R_1 \\ R_2 \\ R_3 \\ \end{array} & \begin{bmatrix} 2 & 1 \\ 3 & 1.5 \\ -4 & -2 \\ \vec{A} & \vec{B} \end{bmatrix}_{3\times 2} \end{array} \xrightarrow{\text{Rank}} \underbrace{\rho = 1}$$

$$\boxed{\begin{array}{c} R_2 = 3/2\, R_1 \\ R_3 = -2R_1 \end{array}}$$

$$\boxed{C_2 = C_1/2}$$

1 I.D. Row

and

1 I.D. Col.

(b) $\vec{A} = 2\hat{i} + \hat{j} - \hat{k}$ $\qquad$ $\vec{A} + \vec{B} = \vec{C}$

$\vec{B} = 3\hat{i} + \hat{j} - 3\hat{k}$

$\vec{C} = 5\hat{i} + 2\hat{j} - 4\hat{k}$

$\underline{2}$ L.I.D. and $\underline{1}$ L.D.

"Col^m-wise" Matrix representation:-

$$\begin{bmatrix} 2 & 3 & 5 \\ 1 & 1 & 2 \\ -1 & -3 & -4 \end{bmatrix} \rightarrow \boxed{\rho = 2}$$

"Row-wise" Matrix representation:-

$$\begin{bmatrix} 2 & 1 & -1 \\ 3 & 1 & -3 \\ 5 & 2 & -4 \end{bmatrix}$$

$R_3 \to R_3 - R_1 - R_2$

$R_2 \to 2R_2 - 3R_1$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & 0 \end{bmatrix} \quad \boxed{\rho = 2} \rightarrow \underbrace{\begin{matrix} 2 \text{ L.I.D.} \\ 1 \text{ L.D.} \end{matrix}}$$

set of L.D. Row

"Col^m-wise" Matrix representation:-

$$\begin{bmatrix} 2 & 3 & 5 \\ 1 & 1 & 2 \\ -1 & -3 & -4 \end{bmatrix} \rightsquigarrow \boxed{\rho = 2} \xrightarrow[R_3 \to R_3 + R_2]{R_1 \to R_1 - 2R_2} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 0 & -2 & -2 \end{bmatrix}$$

Det $\neq 0$

3×3 2×2 1×1

Det $= 0$ $\boxed{\text{Rank} = 2}$

R.E. Method:- $C_3 \to C_3 - (C_1 + C_2)$ ✗ ✗ ✗

2 L.I.D. columns

1 L.D. column

each column represents a vector.

↳ 2 L.I.D. vectors and 1 L.D. vector.

(c) $\vec{A} = 2\hat{i} + \hat{j} - \hat{k}$

$\vec{B} = 3\hat{i} + \hat{j} - 3\hat{k}$

$\vec{C} = 4\hat{i} + 2\hat{j} - 4\hat{k}$

<u>3</u> L.I.D. and <u>0</u> L.D.

"Row-wise" Matrix representation:-

"Column-wise" Matrix representation:-

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 1 & 2 \\ -1 & -3 & -4 \end{bmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_1 \to R_1 - 2R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & -2 & -2 \end{bmatrix}$$

3 L.I.D. columns

0 L.D. columns

$\boxed{\text{3 L.I.D. vectors}}$

$\boxed{\text{Rank} = 3}$

$\underbrace{3\times3}$ $\quad 2\times2 \quad 1\times1$

Det $\neq 0$

Q. Find the value of λ for which following vectors are linear dependent. $x_1 = \begin{bmatrix} 1 \\ 2 \\ \lambda \end{bmatrix}$; $x_2 = \begin{bmatrix} -2 \\ 4 \\ 3 \end{bmatrix}$; $x_3 = \begin{bmatrix} 3 \\ 4 \\ 2\lambda \end{bmatrix}$

4

$$X = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & 4 \\ \lambda & 3 & 2\lambda \end{bmatrix}_{3\times 3}$$

$\underbrace{3\times 3}$ $\quad \underbrace{2\times 2}$ $\quad \underbrace{1\times 1}$

$\downarrow$

Det = 0

For vector being L.D.

$\boxed{\rho(X) < 3 \quad \text{or} \quad \rho(X) \leq 2}$

$\boxed{\rho(X) \neq 3}$

$\downarrow$

$\boxed{|X| = 0}$ *

$$\begin{vmatrix} 1 & -2 & 3 \\ 2 & 4 & 4 \\ \lambda & 3 & 2\lambda \end{vmatrix} = 0$$

$$1(8\lambda - 12) + 2(4\lambda - 4\lambda) + 3(6 - 4\lambda) = 0$$

$$8\lambda - 12 + 18 - 12\lambda = 0$$

$$6 - 4\lambda = 0$$

$$\boxed{\lambda = 1.5}$$ ANS

* Properties of Rank of a Matrix :-

1. $\rho(A) = \rho(A^T) = \rho(A^{-1}) = \rho(A^\theta) = \rho(A \cdot A^T) = \rho(A \cdot A^\theta)$ ✓

2. $[A]_{n\times n}$;

$$\boxed{\begin{array}{l} \rho(A) = n \quad \text{then} \quad \rho(\text{adj}\,A) = n \\ \rho(A) = n-1 \quad \text{then} \quad \rho(\text{adj}\,A) = 1 \\ \rho(A) \le n-2 \quad \text{then} \quad \rho(\text{adj}\,A) = 0 \end{array}}$$

↳ Null Matrix

$[A]_{4\times4}$; $\rho(A) = 4$; $\rho(\text{adj}\,A) = 4$

$\rho(A) = 3$; $\rho(\text{adj}\,A) = 1$

$\rho(A) = 2$ or 1 ; $\rho(\text{adj}\,A) = 0$

$[A]_{4\times4}$: $\rho(A) = 3$

$$A = \begin{bmatrix} a & b & c & d \\ 0 & e & f & g \\ 0 & 0 & h & i \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4\times4}$$

$\rho(A) = 3$

$$\text{adj}(A) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & aeh \end{bmatrix}_{4\times4}$$

Non-zero row

$\boxed{\rho(\text{adj}\,A) = 1}$

Q. $[A]_{5\times5}$; $\rho(A) = 5$; correct statement?

(a) $\rho(A) = \rho(A^T) = \rho(AA^T) = \rho(AA^\theta) = \rho(A^\theta) = \rho(A^{-1}) = 5$

(b) A is non-singular Matrix $\Rightarrow |A| \neq 0$

(c) $\rho[\text{adj} A] = 5$

(d) $\rho[\text{adj}(\text{adj} A)] = 5$

(e) $\rho[\text{adj}(\text{adj} \ldots\ldots \text{adj}(A)] = 5$

$\hookrightarrow$ $\rho(A) = n$; $\rho(\text{adj} A) = n = 5$

$\rho[\text{adj}(\text{adj} A)] \Rightarrow \underset{5\times5}{B} = \underbrace{\text{adj} A}_{5\times5}$; $\underbrace{\rho(B)}_{5\times5} = 5 = n$; $\rho[\text{adj } B] = n = 5$

** $\boxed{\rho[\text{adj}(\text{adj} A)] = 5}$

$\hookrightarrow \quad adj\,(adj\,A) = \underbrace{C}_{5\times 5} \quad ; \quad \rho(C) = 5 \quad ; \quad \rho[adj\,C] = 5$

$$\rho[adj\,(adj\,(adj\,A))] = 5$$

Q. $[A]_{5\times5}$; $\rho(A) = 4$; correct statement?

(a) $\rho(A) = \rho(A^T) = \rho(AA^T) = \rho(AA^\theta) = \rho(A^\theta) = 4$

(b) $|A| = 0$ and A^{-1} doesn't exist.

(c) $\rho[\text{adj}A] = 1$

(d) $\rho[\text{adj}(\text{adj}A)] = 0$

(e) $\rho[\text{adj}(\text{adj} \text{ - - - - -- } \text{adj}(A)] = 0$

↳ If $\rho(A) < n$; then $|A| = 0$

$[A]_{5\times5} \rightarrow \rho(A) = 4 = n-1 \Rightarrow \rho[\text{adj}A] = 1$

$\text{adj } A = [B]_{5\times 5}$

$$[B]_{5\times 5} \longrightarrow \rho(B) = \rho(\text{adj } A) = 1$$

$$\downarrow \qquad (\leq n-2)$$

$$\rho(B) = 1 \leq 5-2$$

$$1 \leq 3$$

$$\rho[\text{adj } B] = 0$$

$$\boxed{\rho[\text{adj}(\text{adj} A)] = 0}$$

$\underbrace{\text{adj}(\text{adj} A)}$ is a null matrix.
$\text{adj}[\text{adj}(\text{adj} A)]$ is " " ".
$\text{adj}[\text{adj} --- \text{adj} A]$ is " " "

$\Big\}$ Rank = 0

Q. $[A]_{n\times n}$ is a singular Non-null matrix and $\rho(A) = \rho(\text{adj}A)$

value of n ?

$\rho(A) \neq 0$

$[A]_{n\times n}$: $\rho(A) = n = \rho(\text{adj}A)$

$\rho(A) = n-1$; $\rho(\text{adj}A) = 1$

$\rho(A) = n-2$; $\rho(\text{adj}A) = 0$

$[A]_{n\times n}$ is singular matrix $\Rightarrow |A| = 0 \Rightarrow \rho(A) < n$ or $\rho(A) \leq n-1$

$\rho(A) \neq 0$

$[A]_{2\times 2}$; $\rho(A) = 1$; $\rho(\text{adj}A) = 1$

$= n-1$

$\boxed{\rho(A) = \rho(\text{adj}A) = 1}$

$\boxed{n = 2}$

Q. $[A]_{n\times n}$ non-null matrix with $\rho(A) = n-4$. Correct statement?

~~(a)~~ $n \geq 5$ ~~(b)~~ Adj A is a null matrix.

~~(c)~~ adj(adj A) could be non-null matrix.

~~(d)~~ $\rho[\text{Adj}(\text{adj} A)] = 0$

↳ $\rho(A) > 0$; $n - 4 > 0 \Rightarrow n > 4$ or $n \geq 5$

adj A : Null Matrix

$[A]_{n\times n}$ and $\rho(A) \leq n-2 \Rightarrow \rho[\text{adj} A] = 0$

$\rho(A) = n-4$

adj [adj A] : Null Matrix

★ Concept of Nullity :-

$$\text{Nullity} = \underbrace{\text{no. of columns}}_{\text{why?}} - \text{Rank}$$

* Some More Properties of Rank :-

(a) $\boxed{\rho(AB) \leq \min[\rho(A), \rho(B)]}$

let $\rho(A) = 2$, $\rho(B) = 4$

then $\rho(AB) \leq \min[2, 4]$; $\rho(AB) \leq 2 \begin{cases} 2 \\ 1 \\ 0 \end{cases}$

(b) $\boxed{\rho(AB) \geq \rho(A) + \rho(B) - n}$ $[A]_{m\times \underline{n}} \cdot [B]_{\underline{n}\times p} = [AB]_{m\times p}$

let $[A]_{3\times \underline{4}}$, $[B]_{\underline{4}\times 5}$

$\rho(A) = 2$ $\rho(B) = 3$ $\rho(AB) \leq \min[2, 3]$

$\boxed{\rho(AB) \leq 2}$

then $\rho(AB) \geq 2 + 3 - 4$

$\boxed{\rho(AB) \geq 1}$

$\therefore \rho(AB) \begin{cases} 1 \\ 2 \end{cases}$

$$A = \begin{bmatrix} a & b & c & d \\ 0 & f & g & h \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4} \; ; \quad B = \begin{bmatrix} p & q & r & s \\ t & u & v & w \\ x & y & z & \alpha \\ \beta & \gamma & \eta & \alpha^* \end{bmatrix}_{4\times 4}$$

$$\rho(A) = 2 \qquad \rho(B) = 4$$

$$AB = \begin{bmatrix} \sim & \sim & \sim & \sim \\ \sim & \sim & \sim & \sim \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4}$$

NON ZERO / ZERO

$$\rho(AB) \to 0,\ 1,\ 2 \Rightarrow \rho(AB) \leq 2$$

$$\rho(AB) \leq \min[2, 4]$$

$$\boxed{\rho(AB) \leq \min[\rho(A), \rho(B)]}$$

$$[A] = \begin{bmatrix} a & b & c & d \\ 0 & \alpha & \beta & \gamma \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4} \; ; \quad [B] = \begin{bmatrix} 0 & C & D & E & F \\ 0 & 0 & G & H & I \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 5}$$

$\rho(A) = 2$

$\rho(B) = 2$

$$AB = \begin{array}{l} \begin{bmatrix} 0 & ac & & & \\ 0 & 0 & \alpha G & & \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 5} \end{array} \begin{array}{l} \to \text{NON-zero Row} \\ \to \text{NON-zero Row} \\ \to \text{zero Row} \end{array}$$

$$\boxed{\rho(AB) = 2}$$

$$\boxed{\rho(AB) \geq 0}$$

$\rho(AB) \geq 2 + 2 - 4$

$\rho(AB) \geq 0$

$$[A] = \begin{bmatrix} 0 & 0 & c & d \\ 0 & 0 & 0 & Y \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4} \; ; \quad [B] = \begin{bmatrix} 0 & C & D & E & F \\ 0 & 0 & G & H & I \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 5}$$

$$\rho(A) = 2 \qquad\qquad \rho(B) = 2$$

$$AB = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\rho(AB) = 0}$$

$$[A] = \begin{bmatrix} 0 & b & c & d \\ 0 & 0 & \beta & \gamma \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4} \quad ; \quad [B] = \begin{bmatrix} 0 & C & D & E & F \\ 0 & 0 & G & H & I \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 5}$$

$$\rho(A) = 2 \qquad\qquad \rho(B) = 2$$

$$AB = \begin{array}{c} \begin{bmatrix} 0 & 0 & bG & \sim & \sim \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \} \to \text{NON-zero} \\ \\ \\ \end{array}$$

$$\boxed{\rho(AB) = 1}$$

$$\boxed{\rho(A)+\rho(B)-n \leq \rho(AB) \leq \min[\rho(A), \rho(B)]}$$

$[A]_{m\times n}$ $[B]_{n\times p}$

common dimension

MSQ

Q. $[A]_{4\times 3}$: $\rho(A) = 2$

$[B]_{5\times 4}$: $\rho(B) = 4$

Possible values of $\rho(BA) = ?$

(a) 1 (b) 2 (c) 3 (d) 4

$\rho(BA) \leq \min[2, 4]$

$$\boxed{\rho(BA) \leq 2}$$

$\rho(BA) \geq \rho(A) + \rho(B) - n$

$\geq 2 + 4 - 4$

$$\boxed{\rho(BA) \geq 2}$$

$$\boxed{\rho(AB) = 2}$$ Ans.

↳ $\boxed{\begin{array}{l} \rho(A+B) \leq \rho(A) + \rho(B) \\ \rho(A-B) \geq \rho(A) - \rho(B) \end{array}}$

$[A]_{4\times4} \rightarrow \rho(A) = 2$

$[B]_{4\times4} \rightarrow \rho(B) = 1$

$\boxed{\rho(A+B) \leq 3}$

$\boxed{\rho(A-B) \geq 1}$

* **System of Linear equations:-**
  - Homogenous ✓
  - Non-Homogenous ✓

(a)
$$2x_1^1 - 5x_2^1 + 3x_3^1 = \underline{0}$$
$$4x_1^1 - 2x_3^1 = \underline{0} \Rightarrow 4x_1^1 + 0 \cdot x_2^1 - 2x_3^1 = 0$$
$$-3x_1^1 + 4x_2^1 - x_3^1 = \underline{0}$$
Homogenous

(b)
$$2x_1 - 5x_2 + 3x_3 = \underline{2}$$
$$4x_1 - 2x_3 = \underline{-4}$$
$$-3x_1 + 4x_2 - x_3 = \underline{0}$$
Non-Homogenous

(c)
$$2x_1 - 4x_2 + 6x_3 = \underline{1}$$
$$3x_1 - 2x_3 = \underline{2}$$
Non-Homogenous

(d)
$$x_1 - 3x_3 = \underline{0}$$
$$x_1 + 2x_2 - 3x_3 = \underline{0}$$
H.

Matrix representation:-

$$\boxed{AX = B}$$

If $B = 0$: Homogenous

If $B \neq 0$: Non. Homogenous

(a) $2x_1 - 5x_2 + 3x_3 = 2$

$4x_1 - 2x_3 = -4 \Rightarrow 4x_1 + 0x_2 - 2x_3 = -4$

$-3x_1 + 4x_2 - x_3 = 1$

$$\underbrace{\begin{bmatrix} 2 & -5 & 3 \\ 4 & 0 & -2 \\ -3 & 4 & -1 \end{bmatrix}_{3\times 3}}_{A} \cdot \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{3\times 1}}_{X} = \underbrace{\begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix}_{3\times 1}}_{B}$$

No. of columns =

(b) $2x_1 - 4x_2 + 6x_3 = 1$

$3x_1 - 2x_3 = 2$

$$\underbrace{\begin{bmatrix} 2 & -4 & 6 \\ 3 & 0 & -2 \end{bmatrix}_{2\times 3}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{3\times 1}}_{X} = \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2\times 1}}_{B}$$

A → co-efficient M

X → variable M

B → const. M.

Solving Non-Homogenous equation :- $Ax = B$

(a) $x_1 + x_2 + x_3 = 6$ — (1)

$2x_1 + 3x_2 + x_3 = 11$ — (2)

$x_1 + 2x_2 + 3x_3 = 14$ — (3)

$$\boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}$$

→ Unique sol^n

[Trivial sol^n]

(2) − 2·(1) → $R_2 - 2R_1$

$x_2 - x_3 = -1$ — (4)

(3) − (1) → $R_3 - R_1$

$x_2 + 2x_3 = 8$ — (5)

(5) − (4) → $R_3 - R_2$

$3x_3 = 9$ — (6)

$$\boxed{\begin{aligned} x_1 + x_2 + x_3 &= 6 \\ x_2 - x_3 &= -1 \\ 3x_3 &= 9 \end{aligned}}$$

$x_3 = 3$, $x_2 = 2$, $x_1 = 1$

Matrix representation:-

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 6 \\ 11 \\ 14 \end{bmatrix}}_{B}$$

Augmented Matrix

$$C = (A:B) = \left[\begin{array}{ccccc} 1 & 1 & 1 & : & 6 \\ 2 & 3 & 1 & : & 11 \\ 1 & 2 & 3 & : & 14 \end{array}\right]$$

with the first three columns labelled A and the last column labelled B.

$$C = A:B = \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 1 & 2 & 3 & 14 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & 2 & 8 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ \underbrace{0 \quad 0 \quad 3}_{A} & & & 9 \end{array}\right] \quad \text{R-E.V}$$

$$\underbrace{}_{A:B}$$

Equivalent

$\rho(A) = 3$

$\rho(A:B) = 3$

$$\boxed{\rho(A) = \rho(A:B) = 3 = \text{Number of variables}}$$

$\downarrow$

unique sol^n

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ \underbrace{0 \quad 0 \quad 3}_{A} & & & \underbrace{9}_{B} \end{array}\right]$$

$$Ax = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \\ 9 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 6$$

$$x_2 - x_3 = -1$$

$$3x_3 = 9 \rightsquigarrow \boxed{x_3 = 3}$$

$$\boxed{x_2 = 2}$$

$$\boxed{x_1 = 1}$$

(b) $x_1 + x_2 + x_3 = 3$ — ①

$2x_1 + 2x_2 + 2x_3 = 6$ — ②

$x_1 - x_2 + x_3 = 1$ — ③

same

$$\left.\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_1 + x_2 + x_3 &= 3 \end{aligned}\right\} — ④$$

(2) − 2(1)

$0 = 0$

(3) − (1)

$-2x_2 = -2$

$\boxed{x_2 = 1}$ ✓

$x_1 + 1 + x_3 = 3$

$\boxed{x_1 + x_3 = 2}$

$x_3 = t$

$x_1 = 2 - t$

$$\boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2-t \\ 1 \\ t \end{bmatrix}}$$

Infinite sol^n

[Non-Trivial] sol^n

↳ $x_1 = 1$, $x_3 = 1$, $x_2 = 1$ ✓

↳ $x_1 = 0$, $x_3 = 2$, $x_2 = 1$ ✓

↳ $x_1 = 2$, $x_3 = 0$, $x_2 = 1$ ✓

↳ $x_1 = -1$, $x_3 = 3$, $x_2 = 1$ ✓

⋮

$\boxed{\text{\# Free variables} = 1}$ → No. of Independent sol^n

Matrix representation:-

variables = 3

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}$$

$$\underbrace{\begin{array}{l} 2 \text{ eq}^n \\ 3 \text{ variables} \end{array}} \to \boxed{\begin{array}{l} x_1 + x_2 + x_3 = 3 \\ -2x_2 = -2 \end{array}}$$

$\Downarrow$

1 free vari.

$$A:B = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 2 & 2 & 2 & : & 6 \\ 1 & -1 & 1 & : & 1 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 0 & 0 & : & 0 \\ 0 & -2 & 0 & : & -2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \underbrace{\begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & -2 & 0 & : & -2 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}}_{A:B}$$

(The first three columns of the last matrix are marked A.)

$$\boxed{\rho(A) = 2, \quad \rho(A:B) = 2}$$

$$\boxed{\rho(A) = \rho(A:B) = 2 < \text{No. of variables}} \to \text{Infinite sol}^n \left[\text{NON-Trivial}\right]$$

(c) $x_1 + x_2 + x_3 = 3$ — ①

$2x_1 + 2x_2 + 2x_3 = 7$ — ②

$x - x_2 + x_3 = 1$ — ③

$\boxed{\text{No sol}^n \text{ Exists}}$

↓

<u>Incosistent solution</u>

$$\boxed{\begin{aligned} x_1 + x_2 + x_3 &= 3 \quad — ④ \\ x_1 + x_2 + x_3 &= 3.5 \quad — ⑤ \end{aligned}}$$

→ possible ?

⑤ - ④

$\boxed{0 = 0.5}$ X X

Matrix representation:-

$$\underset{3\times 3}{\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & -1 & 1 \end{bmatrix}} \underset{3\times 1}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \begin{bmatrix} 3 \\ 7 \\ 1 \end{bmatrix}$$

$$\boxed{0\cdot x_1 + 0\cdot x_2 + 0\cdot x_3 = 1}$$

↳ No Soln

$$A:B = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 2 & 2 & 2 & : & 7 \\ 1 & -1 & 1 & : & 1 \end{bmatrix} \xrightarrow[\substack{R_3 \to R_3 - R_1 \\ \text{Interchange}}]{R_2 \to R_2 - 2R_1} \begin{array}{c} \underbrace{\begin{bmatrix} \underbrace{\begin{matrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{matrix}}_{A} & \begin{matrix} : & 3 \\ : & -2 \\ : & 1 \end{matrix} \end{bmatrix}}_{A:B} \end{array} \begin{array}{l} \to 0 \\ \to 1 \\ \to 3 \end{array}$$

$\rho(A) = 2$, $\rho(A:B) = 3$

$\boxed{\rho(A) \neq \rho(A:B)}$ → No solution

System of Non-Homogenous Linear Equation $[AX=B]$

No. of solⁿ possible
- No solⁿ
- Unique solⁿ
- Infinite solⁿ

Inconsistent solⁿ

No solution Exist

$\rho(A) \neq \rho(A:B)$

Consistent solution

$\rho(A) = \rho(A:B)$

Solution exists

- Infinite solⁿ [NON - Trivial] ✓ → $\rho(A) = \rho(A:B) < n$
- Unique solⁿ [Trivial] ✓ → $\rho(A) = \rho(A:B) = n$

n = no. of Variables

Q. Find the solution and the nature of solution.

$$x_1 + x_2 + x_3 = 3 \quad \text{--- (1)}$$

$$2x_1 + 2x_2 + 2x_3 = 6 \quad \text{--- (2)}$$

$$3x_1 + 3x_2 + 3x_3 = 9 \quad \text{--- (3)}$$

↳ No. of free variables or No of independent solⁿ ?

↳ 3 variables

1 eqⁿ

Free variables = 2

↳ $(1) = (2) = (3)$

$\boxed{x_1 + x_2 + x_3 = 3}$ ↝ Effective eqⁿ

$x_1 = 1, \quad x_2 = 1, \quad x_3 = 1$

$x_1 = 0, \quad x_2 = 1, \quad x_3 = 2$

$x_1 = 2, \quad x_2 = 0, \quad x_3 = 1$

⋮

$x_1 = c_1, \quad x_2 = c_2, \quad x_3 = 3 - c_1 - c_2$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ 3 - c_1 - c_2 \end{bmatrix}$$

$$A:B = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 2 & 2 & 2 & : & 6 \\ 3 & 3 & 3 & : & 9 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \underbrace{\begin{array}{ccccc} \downarrow & \downarrow & \downarrow & & \\ \end{array} \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}}_{A:B}$$

(with the first three columns labelled A)

$\rho(A) = 1$

$\rho(A:B) = 1$

** $\boxed{\text{No. of columns in } A = \text{No. of variables}}$

$\boxed{\rho(A) = \rho(A:B) = 1 < \text{No. of variables}}$

$\Downarrow$

cosistent sol^n, Infinite sol^n, Non. Trivial sol^n

$\eta(A) = \text{No. of variables} - \text{Rank}(A) = 3 - 1 = \boxed{2}$ → No. of free variables

No. of Independent sol^n

↳ Understanding what nullity represent :-

(a) $x_1 + x_2 + x_3 = 3$

$2x_1 + 2x_2 + 2x_3 = 6$

$x_1 - x_2 + x_3 = 1$

variables = 3

↳ sol^n :- $(x_1, x_2, x_3) = (\underset{\text{Dependent}}{2-t}, \underset{\text{fixed}}{1}, \underset{\text{Free variable}}{t})$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 2 & 2 & 2 & : & 6 \\ 1 & -1 & 1 & : & 1 \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & -2 & 0 & : & -2 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = \rho(A:B) = 2 < 3$

$\Downarrow$

Infinite sol^n

Nullity $= 3 - 2 = \underline{\underline{1}}$

$\eta(A) = 1 =$ No. of Free variable or Independent sol^n

(b) $x_1 + x_2 + x_3 = 3$

$2x_1 + 2x_2 + 2x_3 = 6$

$3x_1 + 3x_2 + 3x_3 = 9$

Free variables ... dependent

↳ $\text{sol}^n :- (x_1, x_2, x_3) = (c_1, c_2, 3 - c_1 - c_2)$

$$[A:B] = \left[\begin{array}{ccc:c} 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 6 \\ 3 & 3 & 3 & 9 \end{array}\right] \equiv \left[\begin{array}{ccc:c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\rho(A) = \rho(A:B) = 1$

$\Downarrow$

Infinite sol^n

$$\boxed{\eta(A) = 3 - 1 = 2 = \text{No. of independent sol}^n}$$

Q. Find the number of free variables for the given system.

$$2x + 3y - z = 1 \qquad \# \text{variables} = 3$$
$$3x - 2y + z = 2$$

$$A:B = \left[\begin{array}{ccc:c} 2 & 3 & -1 & 1 \\ 3 & -2 & 1 & 2 \end{array}\right] \xrightarrow{R_2 \to 2R_2 - 3R_1} \left[\begin{array}{ccc:c} 2 & 3 & -1 & 1 \\ 0 & -13 & 5 & 1 \end{array}\right] \text{ REF}$$

$$\rho(A) = 2 \quad , \quad \rho(A:B) = 2$$

$$\underbrace{\rho(A) = \rho(A:B) = 2 < \# \text{variables}}$$

$\Downarrow$

Infinite [Non-Trivial] sol^n

$$\boxed{\eta(A) = 3 - 2 = 1}$$

$\downarrow$

no. of free variables

↳ Is It possible of get $\rho(A) > \rho(A:B) \rightarrow$ NO

$$A:B = \left[\begin{array}{ccc|c} \alpha & \beta & \gamma & \eta \\ 0 & a & b & c \\ 0 & 0 & m & 0 \end{array}\right]$$

$$\underbrace{\underbrace{\begin{matrix} \alpha & \beta & \gamma \\ 0 & a & b \\ 0 & 0 & m \end{matrix}}_{A} \;\; \begin{matrix} \eta \\ c \\ 0 \end{matrix}}_{A:B}$$

$\rho(A) = 3$

$\rho(A:B) = 3$

↳ $\rho(A) = \rho(A:B) \rightarrow$ consistent

↳ $\rho(A) < \rho(A:B) \rightarrow$ Incosistent

↳ $\rho(A) > \rho(A:B) \rightarrow$ NEVER possible

↳ <u>Solving Homogenous equation:-</u>

$\boxed{AX=0}$ → $\underbrace{\rho(A) = \rho(A;B)}_{\text{sol}^n \text{ exist}}$

(a) $x_1 + x_2 + x_3 = 0$ — ①

$2x_1 + 3x_2 + x_3 = 0$ — ②

$x_1 + 2x_2 + 3x_3 = 0$ — ③

$$\boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

↓

$\boxed{\text{unique sol}^n\text{, Trivial sol}^n \quad \text{zero sol}^n}$

(2) − 2(1)

$x_2 - x_3 = 0$ — ④

(3) − (1)

$x_2 + 2x_3 = 0$ — ⑤

↳ zero sol^n ≠ No sol^n d

↓

All the variables are zero.

(5) − (4)

$3x_3 = 0$ → $\boxed{x_3 = 0}$ $\boxed{x_2 = 0}$ $\boxed{x_1 = 0}$

Matrix representation:-

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{O}$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix}}_{\rho(A) = 3}$$

$$\boxed{\rho(A) = 3 = \text{no. of variables} = \text{unique sol}^n}$$

(b) $x_1 + x_2 + x_3 = 0$ — (1)

$2x_1 + 2x_2 + 2x_3 = 0$ — (2)

→ Same

$x_1 - x_2 + x_3 = 0$ — (3)

$x_1 + x_2 + x_3 = 0$ — (4)

$x_1 - x_2 + x_3 = 0$ — (5)

$2x_1 + 2x_3 = 0$

$\boxed{x_1 + x_3 = 0}$

$x_1 + x_2 + x_3 = 0$

$\boxed{x_2 = 0}$

$x_1 = 3, \quad x_2 = 0, \quad x_3 = -3$

$x_1 = -1, \quad x_2 = 0, \quad x_3 = 1$

$x_1 = 9, \quad x_2 = 0, \quad x_3 = -9$

$x_1 = t, \quad x_2 = 0, \quad x_3 = -t$

$$\boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ -t \end{bmatrix}}$$

$\boxed{\#\ \text{Free variables} = 1}$

→ Infinite sol^n

Matrix representation:-

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & -1 & 1 \end{bmatrix} \xrightarrow[\substack{R_3 \to R_3 - R_1 \\ \text{Interchange}}]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\rho(A) = 2 < \text{No. of variables} \Rightarrow \text{Infinite sol}^n}$$

$\eta(A) = 3 - 2$

$= 1$

No. of free variables

System of Homogenous Linear Equation $Ax = 0$

n : no. of variables

= no. of columns in A

Trivial solⁿ

Unique solⁿ

$\boxed{\rho(A) = n}$

$\downarrow$

$\boxed{X = 0}$ $\rightarrow$ <u>zero solⁿ</u>

Non-Trivial solⁿ

Infinite solⁿ

$\boxed{\rho(A) < n}$

$\eta(A)$ = No of variable $- \rho(A)$

= No. of Independent solⁿ